



The octonion linear canonical transform: Properties and applications

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ABSTRACT

The octonion linear canonical transform (OCLCT) is a generalized form of the octonion Fourier transform (OFT), which in recent years has gradually become a new research area at the intersection of mathematics and signal processing. The study of these transforms not only enriches algebraic content but also provides tools for understanding geometric and physical phenomena in higher dimensions. In this work, we study the properties and potential applications of OCLCTs. First, we derive the differential properties and convolution theorem for the left-sided octonion linear canonical transform (LOCLCT). Second, by utilizing the properties and corresponding convolution theorem, we discuss and analyze 3-D linear time-invariant (LTI) systems. Finally, the examples and simulations provided in this study demonstrate the effectiveness of the proposed transform in capturing LOCLCT-frequency components, highlighting its enhanced flexibility and multiscale analysis capabilities.

1. Introduction

The linear canonical transform (LCT) [1] is an important mathematical tool widely used in signal processing [2], optics, and other fields. It is a generalized integral transformation that can be considered a generalization of the FT, fractional Fourier transform [3], generalized Fourier transform, and other transforms [4,5].

The information environment is becoming increasingly complex with the rapid development of information technology, leading to a growing demand for signal processing. To address these challenges, the linear canonical transform (LCT) has gradually expanded its research reach to multidimensional complex domains to address more intricate information processing tasks. Quaternion [6], which is an extended complex number system with four real components, can efficiently represent and process multidimensional signals. They demonstrate great potential in various fields, such as image processing, three-dimensional rotation, and signal processing, through their corresponding transforms, such as the quaternion Fourier transform (QFT) [7], the quaternion fractional Fourier transform (QFRFT) [8,9], the quaternion windowed fractional Fourier transform (QWFRFT) [10], the quaternion linear canonical transform (QLCT) [11–13], the quaternion offset linear canonical transform (QOLCT) [14], and the reduced biquaternion windowed linear canonical transform (RBWLCT) [15]. Octonions, which represent an alternative hypercomplex algebraic structure [16,17], are defined through the Cayley–Dickson construction. In 2021, Gao [18] defined the OCLCT, examined its intrinsic properties and explored the associated uncertainty principle. Later, the concepts of the octonion offset linear canonical transform (OOLCT) and the octonion short-time linear canonical transform (OSTLCT) were generalized by Bhat [19] and Gao [20], respectively. Their basic properties were explored, and the corresponding uncertainty principle was established.

Convolution is a mathematical operation that plays an important role in numerical analysis and signal processing [21,22]. The development of convolution theory provides a solid theoretical foundation for signal processing and differential and integral equations [23,24]. In recent years, significant achievements have been made in the study of convolution theory, particularly in the fields of fractional Fourier transform (FRFT) [25–29] and linear canonical transform (LCT) [30–32]. These studies not only deepen our understanding of convolution theory but also provide crucial theoretical support for practical applications.

The quaternion linear canonical transform (QLCT) is an extension of the LCT in the quaternion field. In recent years, researchers have extensively studied the properties [33] of the QLCT and its applications in signal processing [34]. The octonion is a Cayley–Dickson algebra of order 8, which introduces seven imaginary parts, forming an algebraic structure of higher dimensions than the quaternion. This extension not only increases the

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Table 1
Multiplication rules for octonion algebra.

*	1	e_1	e_2	e_3	e_4	e_5	e_6	e_7
1	1	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	-1	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	e_2	$-e_3$	-1	e_1	e_6	e_7	$-e_4$	$-e_5$
e_3	e_3	e_2	$-e_1$	-1	e_7	$-e_6$	e_5	$-e_4$
e_4	e_4	$-e_5$	$-e_6$	$-e_7$	-1	e_1	e_2	e_3
e_5	e_5	e_4	$-e_7$	e_6	$-e_1$	-1	$-e_3$	e_2
e_6	e_6	e_7	e_4	$-e_5$	$-e_2$	e_3	-1	$-e_1$
e_7	e_7	$-e_6$	e_5	e_4	$-e_3$	$-e_2$	e_1	-1

Table 2
Multiplication rules for $\mathbb{F}$.

*	1	e_1	e_2	e_3	e_4	e_5	e_6	e_7
1	1	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	-1	e_3	$-e_2$	e_5	$-e_4$	e_7	$-e_6$
e_2	e_2	e_3	-1	$-e_1$	e_6	e_7	$-e_4$	$-e_5$
e_3	e_3	$-e_2$	$-e_1$	1	e_7	$-e_6$	$-e_5$	e_4
e_4	e_4	e_5	e_6	e_7	-1	$-e_1$	$-e_2$	$-e_3$
e_5	e_5	$-e_4$	e_7	$-e_6$	$-e_1$	1	$-e_3$	e_2
e_6	e_6	e_7	$-e_4$	$-e_5$	$-e_2$	$-e_3$	1	e_1
e_7	e_7	$-e_6$	$-e_5$	e_4	$-e_3$	e_2	e_1	-1

dimension of the algebra but also introduces new rules and properties for operations, making octonions applicable in a wider range of fields in mathematics.

The OCLCT is a further extension of the LCT in the octonion field. In recent years, significant research achievements have been made in fundamental theories such as convolution [35] and the uncertainty principle [36]. However, the study of the OCLCT is more challenging because of the noncommutative and nonassociative nature of octonion algebra. On the other hand, despite being in its early stages, OCLCT research has potential applications in high-dimensional signal processing, image analysis, and quantum computing. It is therefore necessary and worthwhile to further study the properties of the OCLCT and explore its potential applications to enhance its theoretical framework.

The purpose of this paper is to study the properties and potential applications of the LOCLCT. We first present the definition of the LOCLCT and establish its relationship with the left-sided octonion Fourier transform (LOCFT). Next, we derive various differential properties and a convolution theorem for LOCLCT. By utilizing these properties and the corresponding convolution theorem, we discuss and analyze 3-D LTI. Furthermore, we provide simulations and examples to demonstrate that the proposed transform effectively captures LOCLCT-frequency components with enhanced flexibility and multiscale analysis capabilities. Finally, we analyze and discuss the frequency spectra for LOCLCT while also proposing several potential applications.

The rest of this article is organized as follows. Section 2 provides preliminaries and definitions. In Section 3, we study the differential properties of the LOCLCT. We derive the convolution theorem for the LOCLCT in Section 4. The applications of the convolution theorem to 3-D LTI systems are discussed in Section 5. Two examples for the LOCLCT are given, and simulations are provided to capture the LOCLCT frequency in Section 6. In Section 7, we analyze and discuss the frequency spectra of these two examples. Finally, the conclusions are summarized in Section 8.

2. Preliminaries and definitions

First, we review the basics of octonion algebra and the OCLCT.

2.1. Octonion algebra

Octonion [37] is the algebra of order 8 in the Cayley–Dickson construction, and is noncommutative and nonassociative. An octonion algebra can be expressed as:

$$o = o_0 + o_1 e_1 + o_2 e_2 + o_3 e_3 + o_4 e_4 + o_5 e_5 + o_6 e_6 + o_7 e_7,$$

where $o_0, o_1, \dots, o_7 \in \mathbb{R}$, $e_1, e_2, e_3, e_4, e_5, e_6, e_7$ are imaginary units, and the multiplication rules for octonion algebra are shown in Table 1.

The conjugate and module of an octonion o are defined as:

$$o = o_0 - o_1 e_1 - o_2 e_2 - o_3 e_3 - o_4 e_4 - o_5 e_5 - o_6 e_6 - o_7 e_7,$$

$$|o| = \sqrt{o\bar{o}} = \sqrt{\bar{o}o}.$$

The norm can be defined as $|o|^2 = \sum_{k=0}^7 o_k^2$, and it satisfies the condition $|\varepsilon_1 \varepsilon_2| = |\varepsilon_1| |\varepsilon_2|$ for any $\varepsilon_1, \varepsilon_2 \in \mathbb{O}$.

The octonion-valued function $f(\mathbf{x}) : \mathbb{R}^3 \rightarrow \mathbb{O}$ is defined as:

$$f(\mathbf{x}) = f_0(\mathbf{x}) + f_1(\mathbf{x})e_1 + f_2(\mathbf{x})e_2 + \dots + f_7(\mathbf{x})e_7, \quad (1)$$

where $f_k(\mathbf{x}) \in \mathbb{R}$, $k = 0, 1, \dots, 7$. For the convenience of practical applications, an octonion algebra on the $\mathbb{F}$ field is proposed [36]. The multiplication operation in $\mathbb{F}$ is associative and commutative and its rules follow Table 2.

2.2. Octonion linear canonical transform

Definition 1. For $f \in \mathbb{R}^3$, $M_k = \begin{pmatrix} a_k & b_k \\ c_k & d_k \end{pmatrix}$, $a_k, b_k, c_k, d_k \in \mathbb{R}$, $k = 1, 2, 3$, is a matrix parameter satisfying $\det(M_k) = 1$. The 3-D LOCLCT is defined by

$$\mathcal{L}_l^o\{f\}(\omega) = \int_{\mathbb{R}^3} K f(\mathbf{x}) d\mathbf{x}, \quad (2)$$

where $\mathbf{x} = (x_1, x_2, x_3)$, $\omega = (\omega_1, \omega_2, \omega_3) \in \mathbb{R}^3$. $K = K_{M_1}^{e_1}(x_1, \omega_1) K_{M_2}^{e_2}(x_2, \omega_2) K_{M_3}^{e_4}(x_3, \omega_3)$, and the kernel is defined as:

$$K_{M_1}^{e_1}(x_1, \omega_1) = \frac{1}{\sqrt{2\pi|b_1|}} e^{e_1(\frac{a_1 x_1^2}{2b_1} - \frac{x_1 \omega_1}{b_1} + \frac{d_1 \omega_1^2}{2b_1})}, \quad (3)$$

$$K_{M_2}^{e_2}(x_2, \omega_2) = \frac{1}{\sqrt{2\pi|b_2|}} e^{e_2(\frac{a_2 x_2^2}{2b_2} - \frac{x_2 \omega_2}{b_2} + \frac{d_2 \omega_2^2}{2b_2})}, \quad (4)$$

$$K_{M_3}^{e_4}(x_3, \omega_3) = \frac{1}{\sqrt{2\pi|b_3|}} e^{e_4(\frac{a_3 x_3^2}{2b_3} - \frac{x_3 \omega_3}{b_3} + \frac{d_3 \omega_3^2}{2b_3})}. \quad (5)$$

We are only considering the case where $b_i \neq 0$. For $b_i = 0$, $i = 1, 2, 3$, the LOCLCT of a signal reduces to chirp multiplication, which is not of particular interest for our objective in this work. Hence, unless otherwise specified, we set $b_i \neq 0$ in the following sections.

The inversion formula is as follows:

$$f(\mathbf{x}) = \int_{\mathbb{R}^3} K_{M_3}^{e_4}(\omega_3, x_3) K_{M_2}^{e_2}(\omega_2, x_2) K_{M_1}^{e_1}(\omega_1, x_1) \mathcal{L}_l^o\{f\}(\omega) d\omega, \quad (6)$$

where $M_k^{-1} = \begin{pmatrix} d_k & -b_k \\ -c_k & a_k \end{pmatrix}$, $k = 1, 2, 3$.

Remark 1. When $M_k = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $k = 1, 2, 3$. The LOCLCT reduces to the LOCFT:

$$F_l^o\{f\}(\omega) = \frac{1}{2\pi\sqrt{2\pi}} \int_{\mathbb{R}^3} e^{-e_1 x_1 \omega_1} e^{-e_2 x_2 \omega_2} e^{-e_4 x_3 \omega_3} f(\mathbf{x}) d\mathbf{x}. \quad (7)$$

Remark 2. By shifting the kernel of LOCLCT in Eq. (2) to the right-side of $f(\mathbf{x})$, we can obtain the right-sided OCLCT [19]:

$$\mathcal{L}_l^o\{f\}(\omega) = \int_{\mathbb{R}^3} f(\mathbf{x}) K d\mathbf{x},$$

where

$$K = K_{M_1}^{e_1}(x_1, \omega_1) K_{M_2}^{e_2}(x_2, \omega_2) K_{M_3}^{e_4}(x_3, \omega_3),$$

and the corresponding inversion formula is

$$f(\mathbf{x}) = \int_{\mathbb{R}^3} \mathcal{L}_l^o\{f\}(\omega) K_{M_3}^{e_4}(\omega_3, x_3) K_{M_2}^{e_2}(\omega_2, x_2) K_{M_1}^{e_1}(\omega_1, x_1) d\omega, \quad (8)$$

$$M_k^{-1} = \begin{pmatrix} d_k & -b_k \\ -c_k & a_k \end{pmatrix}, k = 1, 2, 3.$$

3. Differential properties of the left-sided octonion linear canonical transform

In this section, we investigate the properties of the LOCLCT, and obtain a series of differential properties in the time and frequency domains.

Theorem 1. For rapid descent octonion-valued functions f , let $f \in S(\mathbb{R}^3; \mathbb{O})$, the differential properties of 3-D LOCLCT are given as follows:

$$\mathcal{L}_l^o\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = \frac{a_1}{b_1} \widetilde{\mathcal{L}}_l^o\{x_1 f\}(\omega)(-e_1) - \frac{\omega_1}{b_1} \widetilde{\mathcal{L}}_l^o\{f\}(\omega)(-e_1), \quad (9)$$

$$\mathcal{L}_l^o\left\{\frac{\partial f}{\partial x_2}\right\}(\omega) = \frac{a_1}{b_1} \widetilde{\mathcal{L}}_l^o\{x_1 f\}(\omega)(-e_2) - \frac{\omega_1}{b_1} \widetilde{\mathcal{L}}_l^o\{f\}(\omega)(-e_2), \quad (10)$$

$$\mathcal{L}_l^o\left\{\frac{\partial f}{\partial x_3}\right\}(\omega) = \frac{a_1}{b_1} \widetilde{\mathcal{L}}_l^o\{x_1 f\}(\omega)(-e_4) - \frac{\omega_1}{b_1} \widetilde{\mathcal{L}}_l^o\{f\}(\omega)(-e_4), \quad (11)$$

where

$$\begin{aligned} \widetilde{\mathcal{L}}_l^o\{x_1 f\}(\omega) = & \mathcal{L}_{leee}\{x_1 f\}(\omega) - \mathcal{L}_{loee}\{x_1 f\}(\omega)e_1 + \mathcal{L}_{leoe}\{x_1 f\}(\omega)e_2 - \mathcal{L}_{looe}\{x_1 f\}(\omega)e_3 + \mathcal{L}_{leeo}\{x_1 f\}(\omega)e_4 \\ & - \mathcal{L}_{loeo}\{x_1 f\}(\omega)e_5 + \mathcal{L}_{leoo}\{x_1 f\}(\omega)e_6 - \mathcal{L}_{looo}\{x_1 f\}(\omega)e_7, \end{aligned}$$

$$\begin{aligned} \widetilde{\mathcal{L}}_l^o\{f\}(\omega) = & \mathcal{L}_{leee}\{f\}(\omega) - \mathcal{L}_{loee}\{f\}(\omega)e_1 + \mathcal{L}_{leoe}\{f\}(\omega)e_2 - \mathcal{L}_{looe}\{f\}(\omega)e_3 + \mathcal{L}_{leeo}\{f\}(\omega)e_4 \\ & - \mathcal{L}_{loeo}\{f\}(\omega)e_5 + \mathcal{L}_{leoo}\{f\}(\omega)e_6 - \mathcal{L}_{looo}\{f\}(\omega)e_7. \end{aligned}$$

Proof. We first prove Eq. (9): The 3-D LOCLCT of $f : \mathbb{R}^3 \rightarrow \mathbb{O}$ can be represented as [19]:

$$\begin{aligned} \mathcal{L}_l^o\{f\}(\omega) = & \mathcal{L}_{lee}\{f\}(\omega) + \mathcal{L}_{loe}\{f\}(\omega)e_1 + \mathcal{L}_{leo}\{f\}(\omega)e_2 + \mathcal{L}_{loo}\{f\}(\omega)e_3 + \mathcal{L}_{leeo}\{f\}(\omega)e_4 \\ & + \mathcal{L}_{loeo}\{f\}(\omega)e_5 + \mathcal{L}_{leoo}\{f\}(\omega)e_6 + \mathcal{L}_{looo}\{f\}(\omega)e_7, \end{aligned} \quad (12)$$

where

$$\mathcal{L}_{lee}\{f\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \cos \gamma_1 \cos \gamma_2 \cos \gamma_3 f_{eee}(\mathbf{x}) d\mathbf{x}, \quad (13)$$

$$\mathcal{L}_{loe}\{f\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \sin \gamma_1 \cos \gamma_2 \cos \gamma_3 f_{oeo}(\mathbf{x}) d\mathbf{x}, \quad (14)$$

$$\mathcal{L}_{leo}\{f\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \cos \gamma_1 \sin \gamma_2 \cos \gamma_3 f_{eoe}(\mathbf{x}) d\mathbf{x}, \quad (15)$$

$$\mathcal{L}_{loo}\{f\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \sin \gamma_1 \sin \gamma_2 \cos \gamma_3 f_{ooo}(\mathbf{x}) d\mathbf{x}, \quad (16)$$

$$\mathcal{L}_{leeo}\{f\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \cos \gamma_1 \cos \gamma_2 \sin \gamma_3 f_{eeo}(\mathbf{x}) d\mathbf{x}, \quad (17)$$

$$\mathcal{L}_{loeo}\{f\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \sin \gamma_1 \cos \gamma_2 \sin \gamma_3 f_{oeo}(\mathbf{x}) d\mathbf{x}, \quad (18)$$

$$\mathcal{L}_{leoo}\{f\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \cos \gamma_1 \sin \gamma_2 \sin \gamma_3 f_{eoo}(\mathbf{x}) d\mathbf{x}, \quad (19)$$

$$\mathcal{L}_{looo}\{f\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \sin \gamma_1 \sin \gamma_2 \sin \gamma_3 f_{ooo}(\mathbf{x}) d\mathbf{x}, \quad (20)$$

and $\gamma_i = \frac{a_i}{2b_i}x_i^2 - \frac{x_i\omega_i}{b_i} + \frac{d_i}{2b_i}\omega_i^2, i = 1, 2, 3$. $D_\gamma = \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}}$. Then, $\mathcal{L}_l^o\{\frac{\partial f}{\partial x_1}\}(\omega)$ can be decomposed into the following form:

$$\begin{aligned} \mathcal{L}_l^o\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = & \mathcal{L}_{lee}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) + \mathcal{L}_{loe}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega)e_1 + \mathcal{L}_{leo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega)e_2 + \mathcal{L}_{loo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega)e_3 \\ & + \mathcal{L}_{leeo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega)e_4 + \mathcal{L}_{loeo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega)e_5 + \mathcal{L}_{leoo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega)e_6 + \mathcal{L}_{looo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega)e_7, \end{aligned}$$

where

$$\mathcal{L}_{lee}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \cos \gamma_1 \cos \gamma_2 \cos \gamma_3 \frac{\partial f(\mathbf{x})}{\partial x_1} d\mathbf{x}, \quad (21)$$

$$\mathcal{L}_{loe}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \sin \gamma_1 \cos \gamma_2 \cos \gamma_3 \frac{\partial f(\mathbf{x})}{\partial x_1} d\mathbf{x}, \quad (22)$$

$$\mathcal{L}_{leo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \cos \gamma_1 \sin \gamma_2 \cos \gamma_3 \frac{\partial f(\mathbf{x})}{\partial x_1} d\mathbf{x}, \quad (23)$$

$$\mathcal{L}_{loo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \sin \gamma_1 \sin \gamma_2 \cos \gamma_3 \frac{\partial f(\mathbf{x})}{\partial x_1} d\mathbf{x}, \quad (24)$$

$$\mathcal{L}_{leeo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \cos \gamma_1 \cos \gamma_2 \sin \gamma_3 \frac{\partial f(\mathbf{x})}{\partial x_1} d\mathbf{x}, \quad (25)$$

$$\mathcal{L}_{loeo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \sin \gamma_1 \cos \gamma_2 \sin \gamma_3 \frac{\partial f(\mathbf{x})}{\partial x_1} d\mathbf{x}, \quad (26)$$

$$\mathcal{L}_{leoo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \cos \gamma_1 \sin \gamma_2 \sin \gamma_3 \frac{\partial f(\mathbf{x})}{\partial x_1} d\mathbf{x}, \quad (27)$$

$$\mathcal{L}_{looo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = D_\gamma \int_{\mathbb{R}^3} \sin \gamma_1 \sin \gamma_2 \sin \gamma_3 \frac{\partial f(\mathbf{x})}{\partial x_1} d\mathbf{x}. \quad (28)$$

For every integrable function $f \in S(\mathbb{R}^3; \mathbb{O})$ and $x_2, x_3 \in \mathbb{R}$, we have $\lim_{x_1 \rightarrow \pm\infty} f(\mathbf{x}) = 0$. The results of Eqs. (21)–(28) are as follows:

$$\mathcal{L}_{lee}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = \frac{a_1}{b_1}\mathcal{L}_{loe}\{x_1 f\}(\omega) - \frac{\omega_1}{b_1}\mathcal{L}_{loe}\{f\}(\omega), \quad (29)$$

$$\mathcal{L}_{loe}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = -\left(\frac{a_1}{b_1}\mathcal{L}_{lee}\{x_1 f\}(\omega) - \frac{\omega_1}{b_1}\mathcal{L}_{lee}\{f\}(\omega)\right), \quad (30)$$

$$\mathcal{L}_{leo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = \frac{a_1}{b_1}\mathcal{L}_{loeo}\{x_1 f\}(\omega) - \frac{\omega_1}{b_1}\mathcal{L}_{loeo}\{f\}(\omega), \quad (31)$$

$$\mathcal{L}_{loo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = -\left(\frac{a_1}{b_1}\mathcal{L}_{leo}\{x_1 f\}(\omega) - \frac{\omega_1}{b_1}\mathcal{L}_{leo}\{f\}(\omega)\right), \quad (32)$$

$$\mathcal{L}_{leeo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = \frac{a_1}{b_1}\mathcal{L}_{leoo}\{x_1 f\}(\omega) - \frac{\omega_1}{b_1}\mathcal{L}_{leoo}\{f\}(\omega), \quad (33)$$

$$\mathcal{L}_{loeo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = -\left(\frac{a_1}{b_1}\mathcal{L}_{leeo}\{x_1 f\}(\omega) - \frac{\omega_1}{b_1}\mathcal{L}_{leeo}\{f\}(\omega)\right), \quad (34)$$

$$\mathcal{L}_{leoo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = \frac{a_1}{b_1}\mathcal{L}_{looo}\{x_1 f\}(\omega) - \frac{\omega_1}{b_1}\mathcal{L}_{looo}\{f\}(\omega), \quad (35)$$

$$\mathcal{L}_{loeo}\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) = -\left(\frac{a_1}{b_1}\mathcal{L}_{leeo}\{x_1 f\}(\omega) - \frac{\omega_1}{b_1}\mathcal{L}_{leeo}\{f\}(\omega)\right). \quad (36)$$

From Eq. (12), we have

$$\begin{aligned} & \mathcal{L}_l^o\left\{\frac{\partial f}{\partial x_1}\right\}(\omega) \\ &= \frac{a_1}{b_1} \left[\mathcal{L}_{loee}\{x_1 f\}(\omega) - \mathcal{L}_{leee}\{x_1 f\}(\omega)e_1 + \mathcal{L}_{looe}\{x_1 f\}(\omega)e_2 - \mathcal{L}_{leoe}\{x_1 f\}(\omega)e_3 + \mathcal{L}_{loeo}\{x_1 f\}(\omega)e_4 \right. \\ & \quad \left. - \mathcal{L}_{leeo}\{x_1 f\}(\omega)e_5 + \mathcal{L}_{looo}\{x_1 f\}(\omega)e_6 - \mathcal{L}_{leoo}\{x_1 f\}(\omega)e_7 \right] \\ & \quad - \frac{\omega_1}{b_1} \left[\mathcal{L}_{loee}\{f\}(\omega) - \mathcal{L}_{leee}\{f\}(\omega)e_1 + \mathcal{L}_{looe}\{f\}(\omega)e_2 - \mathcal{L}_{leoe}\{f\}(\omega)e_3 + \mathcal{L}_{loeo}\{f\}(\omega)e_4 - \mathcal{L}_{leeo}\{f\}(\omega)e_5 \right. \\ & \quad \left. + \mathcal{L}_{looo}\{f\}(\omega)e_6 - \mathcal{L}_{leoo}\{f\}(\omega)e_7 \right] \\ &= \frac{a_1}{b_1} \left[\mathcal{L}_{leee}\{x_1 f\}(\omega) - \mathcal{L}_{loee}\{x_1 f\}(\omega)e_1 + \mathcal{L}_{leoe}\{x_1 f\}(\omega)e_2 - \mathcal{L}_{looe}\{x_1 f\}(\omega)e_3 + \mathcal{L}_{leeo}\{x_1 f\}(\omega)e_4 \right. \\ & \quad \left. - \mathcal{L}_{leeo}\{x_1 f\}(\omega)e_5 + \mathcal{L}_{leoo}\{x_1 f\}(\omega)e_6 - \mathcal{L}_{looo}\{x_1 f\}(\omega)e_7 \right] (-e_1) \\ & \quad - \frac{\omega_1}{b_1} \left[\mathcal{L}_{leee}\{f\}(\omega) - \mathcal{L}_{loee}\{f\}(\omega)e_1 + \mathcal{L}_{leoe}\{f\}(\omega)e_2 - \mathcal{L}_{looe}\{f\}(\omega)e_3 + \mathcal{L}_{leeo}\{f\}(\omega)e_4 \right. \\ & \quad \left. - \mathcal{L}_{leeo}\{f\}(\omega)e_5 + \mathcal{L}_{leoo}\{f\}(\omega)e_6 - \mathcal{L}_{looo}\{f\}(\omega)e_7 \right] (-e_1), \end{aligned}$$

which completes the proof of Eq. (9). Similarly, we can prove Eqs. (10) and (11), therefore, we omit their proofs.

Remark 3. We use e or o to denote that the function is even (e) or odd (o) with respect to the proper variable; the function $f_{leeo}(\mathbf{x})$ is even with respect to x_1, x_3 and odd with respect to x_2 .

Corollary 1. Let $\mathcal{L}_l^o\left\{\frac{\partial^k f}{\partial x_1 \dots x_j}\right\}(\omega)$ denote the 3-D LOCLCT of $\frac{\partial^k f}{\partial x_1 \dots x_j}$, $k = 2, 3$. Then, we obtain

$$\begin{aligned} \mathcal{L}_l^o\left\{\frac{\partial^2 f}{\partial x_1 \partial x_2}\right\}(\omega) &= -\frac{a_1 a_2}{b_1 b_2} \mathcal{L}_l^o\{x_1 x_2 f\}(\omega) + \frac{a_1 \omega_2}{b_1 b_2} \{x_1 f\}(\omega) + \frac{a_2 \omega_1}{b_1 b_2} \{x_2 f\}(\omega) - \frac{\omega_1 \omega_2}{b_1 b_2} \{f\}(\omega), \\ \mathcal{L}_l^o\left\{\frac{\partial^2 f}{\partial x_1 \partial x_3}\right\}(\omega) &= -\frac{a_1 a_3}{b_1 b_3} \mathcal{L}_l^o\{x_1 x_3 f\}(\omega) + \frac{a_1 \omega_3}{b_1 b_3} \{x_1 f\}(\omega) + \frac{a_3 \omega_1}{b_1 b_3} \{x_3 f\}(\omega) - \frac{\omega_1 \omega_3}{b_1 b_3} \{f\}(\omega), \\ \mathcal{L}_l^o\left\{\frac{\partial^2 f}{\partial x_2 \partial x_3}\right\}(\omega) &= -\frac{a_2 a_3}{b_2 b_3} \mathcal{L}_l^o\{x_2 x_3 f\}(\omega) + \frac{a_2 \omega_3}{b_2 b_3} \{x_2 f\}(\omega) + \frac{a_3 \omega_2}{b_2 b_3} \{x_3 f\}(\omega) - \frac{\omega_2 \omega_3}{b_2 b_3} \{f\}(\omega), \\ \mathcal{L}_l^o\left\{\frac{\partial^3 f}{\partial x_1 \partial x_2 \partial x_3}\right\}(\omega) &= -\frac{a_1 a_2 a_3}{b_1 b_2 b_3} \mathcal{L}_l^o\{x_1 x_2 x_3 f\}(\omega) + \frac{a_1 a_2 \omega_3}{b_1 b_2 b_3} \{x_1 x_2 f\}(\omega) + \frac{a_1 a_3 \omega_2}{b_1 b_2 b_3} \{x_1 x_3 f\}(\omega) \\ & \quad + \frac{a_2 a_3 \omega_1}{b_1 b_2 b_3} \{x_2 x_3 f\}(\omega) - \frac{a_1 \omega_2 \omega_3}{b_1 b_2 b_3} \{x_1 f\}(\omega) - \frac{a_2 \omega_1 \omega_3}{b_1 b_2 b_3} \{x_2 f\}(\omega) \\ & \quad - \frac{a_3 \omega_1 \omega_2}{b_1 b_2 b_3} \{x_3 f\}(\omega) + \frac{\omega_1 \omega_2 \omega_3}{b_1 b_2 b_3} \{f\}(\omega). \end{aligned}$$

Corollary 2. Let $\mathcal{L}_l^o\left\{\frac{\partial^2 f}{\partial x_i^2}\right\}(\omega)$ denote the 3-D LOCLCT of $\frac{\partial^2 f}{\partial x_i^2}$. Then, we obtain

$$\begin{aligned} \mathcal{L}_l^o\left\{\frac{\partial^2 f}{\partial x_1^2}\right\}(\omega) &= -\frac{a_1^2}{b_1} \mathcal{L}_l^o\{x_1^2 f\}(\omega) + \frac{2a_1 \omega_1}{b_1^2} \{x_1 f\}(\omega) - \frac{\omega_1^2}{b_1} \{f\}(\omega), \\ \mathcal{L}_l^o\left\{\frac{\partial^2 f}{\partial x_2^2}\right\}(\omega) &= -\frac{a_2^2}{b_2} \mathcal{L}_l^o\{x_2^2 f\}(\omega) + \frac{2a_2 \omega_2}{b_2^2} \{x_2 f\}(\omega) - \frac{\omega_2^2}{b_2} \{f\}(\omega), \\ \mathcal{L}_l^o\left\{\frac{\partial^2 f}{\partial x_3^2}\right\}(\omega) &= -\frac{a_3^2}{b_3} \mathcal{L}_l^o\{x_3^2 f\}(\omega) + \frac{2a_3 \omega_3}{b_3^2} \{x_3 f\}(\omega) - \frac{\omega_3^2}{b_3} \{f\}(\omega). \end{aligned}$$

Theorem 2. Let $f \in S(\mathbb{R}^3; \mathbb{O})$. The differential properties of 3-D LOCLCT in the frequency domain are as follows:

$$\frac{\partial \{\mathcal{L}_l^o\{f\}(\omega)\}}{\partial \omega_1} = -\frac{e_1}{b_1} \mathcal{L}_l^o\{x_1 f\}(\omega) + \frac{e_1 d_1 \omega_1}{b_1} \mathcal{L}_l^o\{f\}(\omega), \quad (37)$$

$$\frac{\partial \{\mathcal{L}_l^o\{f\}(\omega)\}}{\partial \omega_3} = -\frac{e_4}{b_3} \mathcal{L}_l^o\{x_3 f\}(\omega) + \frac{e_4 d_3 \omega_3}{b_3} \mathcal{L}_l^o\{f\}(\omega). \quad (38)$$

Proof. First, we prove Eq. (37). From Eq. (2), we have

$$\begin{aligned} \frac{\partial \{\mathcal{L}_l^o\{f\}(\omega)\}}{\partial \omega_1} &= \int_{\mathbb{R}^3} \frac{\partial K_{M_1}^{e_1}(x_1, \omega_1)}{\partial \omega_1} K_{M_2}^{e_2}(x_2, \omega_2) K_{M_3}^{e_4}(x_3, \omega_3) f(\mathbf{x}) d\mathbf{x} \\ &= \int_{\mathbb{R}^3} \frac{1}{\sqrt{2\pi|b_1|}} e_1 \left(\frac{a_1 x_1^2}{2b_1} - \frac{x_1 \omega_1}{b_1} + \frac{d_1 \omega_1^2}{2b_1} \right) e_1 \left(\frac{-x_1 + d_1 \omega_1}{b_1} \right) \\ &\quad \cdot K_{M_2}^{e_2}(x_2, \omega_2) K_{M_3}^{e_4}(x_3, \omega_3) f(\mathbf{x}) d\mathbf{x} \\ &= -\frac{e_1}{b_1} \mathcal{L}_l^o\{x_1 f\}(\omega) + \frac{e_1 d_1 \omega_1}{b_1} \mathcal{L}_l^o\{f\}(\omega), \end{aligned} \quad (39)$$

which completes the proof of Eq. (37). Similarly, the proof of Eq. (38) is omitted.

Theorem 3. Let $f \in S(\mathbb{R}^3; \mathbb{O})$, $\Delta_{x_k} f = \frac{\partial f}{\partial x_k} - \frac{a_k}{b_k} e_q x_k f$, $k = 1, 2, 3$. where e_q denotes the octonion imaginary unit, $q = 1, 2, 4$. For any $p \in \mathbb{Z}^+$, we obtain

$$\Delta_{x_k}^p K_{M_k}(x_k, \omega_k) = \left(-\frac{e_q \omega_k}{b_k}\right)^p K_{M_k}(x_k, \omega_k).$$

Proof. For $p = 1$, we have

$$\begin{aligned} \Delta_{x_k} K_{M_k}(x_k, \omega_k) &= \frac{\partial K_{M_k}(x_k, \omega_k)}{\partial x_k} - \frac{a_k}{b_k} e_q x_k K_{M_k}(x_k, \omega_k) \\ &= \frac{e_q}{b_k} (a_k x_k - \omega_k) K_{M_k}(x_k, \omega_k) - \frac{a_k}{b_k} e_q x_k K_{M_k}(x_k, \omega_k) \\ &= \left(-\frac{e_q \omega_k}{b_k}\right) K_{M_k}(x_k, \omega_k). \end{aligned}$$

Supposing that $\Delta_{x_k}^{p-1} K_{M_k}(x_k, \omega_k) = \left(-\frac{e_q \omega_k}{b_k}\right)^{p-1} K_{M_k}(x_k, \omega_k)$, we have

$$\begin{aligned} \Delta_{x_k}^p K_{M_k}(x_k, \omega_k) &= \Delta_{x_k} [\Delta_{x_k}^{p-1} K_{M_k}(x_k, \omega_k)] \\ &= \frac{\partial \left(-\frac{e_q \omega_k}{b_k}\right)^{p-1} K_{M_k}(x_k, \omega_k)}{\partial x_k} - \frac{a_k}{b_k} e_q x_k \left(-\frac{e_q \omega_k}{b_k}\right)^{p-1} K_{M_k}(x_k, \omega_k) \\ &= \left(-\frac{e_q \omega_k}{b_k}\right)^{p-1} \left[\frac{e_q}{b_k} (a_k x_k - \omega_k) K_{M_k}(x_k, \omega_k) \right] - \frac{a_k}{b_k} e_q x_k \left(-\frac{e_q \omega_k}{b_k}\right)^{p-1} K_{M_k}(x_k, \omega_k) \\ &= \left(-\frac{e_q \omega_k}{b_k}\right)^p K_{M_k}(x_k, \omega_k). \end{aligned}$$

Theorem 4. Let $f \in L^2(\mathbb{R}^3, \mathbb{O})$, $\Delta_{\omega_k} f = \frac{\partial f}{\partial \omega_k} - \frac{d_k}{b_k} e_q \omega_k f$, $k = 1, 2, 3$. For any $p \in \mathbb{Z}^+$, we have

$$\Delta_{\omega_k}^p \{\mathcal{L}_l^o\{f\}(\omega)\} = \mathcal{L}_l^o\left\{\left(-\frac{e_q x_k}{b_k}\right)^p f\right\}(\omega).$$

Proof. For $p = 1$,

$$\begin{aligned} \Delta_{\omega_k} \{\mathcal{L}_l^o\{f\}(\omega)\} &= \frac{\partial \mathcal{L}_l^o\{f\}(\omega)}{\partial \omega_k} - \frac{d_k}{b_k} e_q \omega_k \mathcal{L}_l^o\{f\}(\omega) \\ &= e_q \left(\frac{-x_k + d_k \omega_k}{b_k} \right) \mathcal{L}_l^o\{f\}(\omega) - \frac{d_k}{b_k} e_q \omega_k \mathcal{L}_l^o\{f\}(\omega) \\ &= \mathcal{L}_l^o\left\{\left(-\frac{e_q x_k}{b_k}\right) f\right\}(\omega). \end{aligned}$$

Supposing that $\Delta_{\omega_k}^{p-1} \{\mathcal{L}_l^o\{f\}(\omega)\} = \mathcal{L}_l^o\left\{\left(-\frac{e_q x_k}{b_k}\right)^{p-1} f\right\}(\omega)$, we obtain

$$\begin{aligned} \Delta_{\omega_k}^p \{\mathcal{L}_l^o\{f\}(\omega)\} &= \Delta_{\omega_k} \left\{ \Delta_{\omega_k}^{p-1} \{\mathcal{L}_l^o\{f\}(\omega)\} \right\} \\ &= \Delta_{\omega_k} \left\{ \mathcal{L}_l^o\left\{\left(-\frac{e_q x_k}{b_k}\right)^{p-1} f\right\}(\omega) \right\} \\ &= \frac{\partial \mathcal{L}_l^o\left\{\left(-\frac{e_q x_k}{b_k}\right)^{p-1} f\right\}(\omega)}{\partial \omega_k} - \frac{d_k}{b_k} e_q \omega_k \mathcal{L}_l^o\left\{\left(-\frac{e_q x_k}{b_k}\right)^{p-1} f\right\}(\omega) \\ &= \left(-\frac{e_q x_k}{b_k}\right)^{p-1} \cdot e_q \left(\frac{-x_k + d_k \omega_k}{b_k} \right) \mathcal{L}_l^o\{f\}(\omega) - \frac{d_k}{b_k} e_q \omega_k \left(-\frac{e_q x_k}{b_k}\right)^{p-1} \mathcal{L}_l^o\{f\}(\omega) \\ &= \mathcal{L}_l^o\left\{\left(-\frac{e_q x_k}{b_k}\right)^p f\right\}(\omega). \end{aligned}$$

This completes the proof.

4. Convolution theorem of the left-sided octonion linear canonical transform

In this section, a new convolution operator is proposed, and the corresponding convolution theorem for the 3-D LOCLCT is derived.

Definition 2. Let $f, g \in L^2(\mathbb{R}^3, \mathbb{O})$, a new convolution operator Θ is defined by

$$(f\Theta g)(x) = \int_{\mathbb{R}^3} f(\sigma)g(\mathbf{x}-\sigma) \frac{1}{\sqrt{2\pi b_1}} e^{-e_1(\sigma_1(x_1-\sigma_1))\frac{a_1}{b_1}} \frac{1}{\sqrt{2\pi b_2}} e^{-e_2(\sigma_2(x_2-\sigma_2))\frac{a_2}{b_2}} \frac{1}{\sqrt{2\pi b_3}} e^{-e_4(\sigma_3(x_3-\sigma_3))\frac{a_3}{b_3}} d\sigma.$$

Theorem 5 (Convolution Theorem). Let $\mathcal{L}_l^o\{f\Theta g\}(\omega)$ denote the 3-D LOCLCT of $f\Theta g$. Then, we obtain

$$\begin{aligned} \mathcal{L}_l^o\{f\Theta g\}(\omega) = & e^{-e_1\frac{d_1}{2b_1}\omega_1^2} e^{-e_2\frac{d_2}{2b_2}\omega_2^2} e^{-e_4\frac{d_3}{2b_3}\omega_3^2} \{ \mathcal{L}_l^o\{f_0\}(\omega)\mathcal{L}_l^o\{g\}(\omega) + \mathcal{L}_l^o\{f_1e_1\}(\omega)\mathcal{L}_l^o\{g\}(\omega) \\ & + \mathcal{L}_l^o\{f_2e_2\}(\omega)\mathcal{L}_l^o\{g\}(\omega) + \mathcal{L}_l^o\{f_3e_1e_2\}(\omega)\mathcal{L}_l^o\{g\}(\omega) + \mathcal{L}_l^o\{f_4e_4\}(\omega)\mathcal{L}_l^o\{g\}(\omega) \\ & + \mathcal{L}_l^o\{f_5e_1e_4\}(\omega)\mathcal{L}_l^o\{g\}(\omega) + \mathcal{L}_l^o\{f_6e_2e_4\}(\omega)\mathcal{L}_l^o\{g\}(\omega) + \mathcal{L}_l^o\{f_7e_1e_2e_4\}(\omega)\mathcal{L}_l^o\{g\}(\omega) \}. \end{aligned}$$

Proof. According to the 3-D LOCLCT, we have

$$\begin{aligned} & \mathcal{L}_l^o\{f\Theta g\}(\omega) \\ = & \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} Kf(\sigma)g(\mathbf{x}-\sigma) \frac{1}{\sqrt{2\pi b_1}} e^{-e_1(\sigma_1(x_1-\sigma_1))\frac{a_1}{b_1}} \frac{1}{\sqrt{2\pi b_2}} e^{-e_2(\sigma_2(x_2-\sigma_2))\frac{a_2}{b_2}} \\ & \frac{1}{\sqrt{2\pi b_3}} e^{-e_4(\sigma_3(x_3-\sigma_3))\frac{a_3}{b_3}} d\sigma d\mathbf{x} \\ = & D_\gamma \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} e^{e_1(\frac{a_1x_1^2}{2b_1} - \frac{x_1\omega_1}{b_1} + \frac{d_1}{2b_1}\omega_1^2)} e^{e_2(\frac{a_2x_2^2}{2b_2} - \frac{x_2\omega_2}{b_2} + \frac{d_2}{2b_2}\omega_2^2)} \\ & \cdot e^{e_4(\frac{a_3x_3^2}{2b_3} - \frac{x_3\omega_3}{b_3} + \frac{d_3}{2b_3}\omega_3^2)} f(\sigma)g(\mathbf{x}-\sigma) e^{-e_1(\sigma_1(x_1-\sigma_1))\frac{a_1}{b_1}} e^{-e_2(\sigma_2(x_2-\sigma_2))\frac{a_2}{b_2}} e^{-e_4(\sigma_3(x_3-\sigma_3))\frac{a_3}{b_3}} d\sigma d\mathbf{x}, \end{aligned}$$

where $\sigma = (\sigma_1, \sigma_2, \sigma_3)$, $\mathbf{x} - \sigma = (x_1 - \sigma_1, x_2 - \sigma_2, x_3 - \sigma_3)$.

Letting $s_1 = x_1 - \sigma_1$, $s_2 = x_2 - \sigma_2$, $s_3 = x_3 - \sigma_3$, we obtain

$$\begin{aligned} \mathcal{L}_l^o\{f\Theta g\}(\omega) = & D_\gamma \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} e^{e_1(\frac{a_1(s_1+\sigma_1)^2}{2b_1} - \frac{(s_1+\sigma_1)\omega_1}{b_1} + \frac{d_1}{2b_1}\omega_1^2)} e^{e_2(\frac{a_2(s_2+\sigma_2)^2}{2b_2} - \frac{(s_2+\sigma_2)\omega_2}{b_2} + \frac{d_2}{2b_2}\omega_2^2)} \\ & \cdot e^{e_4(\frac{a_3(s_3+\sigma_3)^2}{2b_3} - \frac{(s_3+\sigma_3)\omega_3}{b_3} + \frac{d_3}{2b_3}\omega_3^2)} f(\sigma)g(s) e^{-e_1(\sigma_1s_1)\frac{a_1}{b_1}} e^{-e_2(\sigma_2s_2)\frac{a_2}{b_2}} e^{-e_4(\sigma_3s_3)\frac{a_3}{b_3}} d\sigma ds. \end{aligned}$$

From Eq. (1), we represent the form of an octonion function as follows:

$$\begin{aligned} f = & f_0 + f_1e_1 + f_2e_2 + f_3e_3 + f_4e_4 + f_5e_5 + f_6e_6 + f_7e_7 \\ = & (f_0 + f_1e_1) + [(f_2 + f_3e_1)e_2] + [(f_4 + f_5e_1)e_4] + [(f_6 + f_7e_1)e_2e_4], \end{aligned}$$

where $f_0 + f_1e_1, f_2 + f_3e_1, f_4 + f_5e_1, f_6 + f_7e_1 \in \mathbb{C}$. Then,

$$\begin{aligned} \mathcal{L}_l^o\{f\Theta g\}(\omega) = & D_\gamma \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} e^{e_1(\frac{a_1(s_1+\sigma_1)^2}{2b_1} - \frac{(s_1+\sigma_1)\omega_1}{b_1} + \frac{d_1}{2b_1}\omega_1^2)} \\ & \cdot e^{e_2(\frac{a_2(s_2+\sigma_2)^2}{2b_2} - \frac{(s_2+\sigma_2)\omega_2}{b_2} + \frac{d_2}{2b_2}\omega_2^2)} e^{e_4(\frac{a_3(s_3+\sigma_3)^2}{2b_3} - \frac{(s_3+\sigma_3)\omega_3}{b_3} + \frac{d_3}{2b_3}\omega_3^2)} \\ & \cdot f(\sigma)g(s) e^{-e_1(\sigma_1s_1)\frac{a_1}{b_1}} e^{-e_2(\sigma_2s_2)\frac{a_2}{b_2}} e^{-e_4(\sigma_3s_3)\frac{a_3}{b_3}} d\sigma ds \\ = & D_\gamma \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} e^{e_1(\frac{a_1(s_1+\sigma_1)^2}{2b_1} - \frac{(s_1+\sigma_1)\omega_1}{b_1} + \frac{d_1}{2b_1}\omega_1^2)} \\ & \cdot e^{e_2(\frac{a_2(s_2+\sigma_2)^2}{2b_2} - \frac{(s_2+\sigma_2)\omega_2}{b_2} + \frac{d_2}{2b_2}\omega_2^2)} e^{e_4(\frac{a_3(s_3+\sigma_3)^2}{2b_3} - \frac{(s_3+\sigma_3)\omega_3}{b_3} + \frac{d_3}{2b_3}\omega_3^2)} \\ & \cdot (f_0 + f_1e_1 + f_2e_2 + f_3e_1e_2 + f_4e_4 + f_5e_1e_4 + f_6e_2e_4 + f_7e_1e_2e_4) g d\sigma ds \\ & \cdot e^{-e_1(\sigma_1s_1)\frac{a_1}{b_1}} e^{-e_2(\sigma_2s_2)\frac{a_2}{b_2}} e^{-e_4(\sigma_3s_3)\frac{a_3}{b_3}} \\ = & e^{-\frac{e_1d_1\omega_1^2}{2b_1}} e^{-\frac{e_2d_2\omega_2^2}{2b_2}} e^{-\frac{e_4d_3\omega_3^2}{2b_3}} (\mathcal{L}\{f_0\}(\omega)\mathcal{L}\{g\}(\omega) + \mathcal{L}_l^o\{f_1e_1\}(\omega)\mathcal{L}_l^o\{g\}(\omega) \\ & + \mathcal{L}_l^o\{f_2e_2\}(\omega)\mathcal{L}_l^o\{g\}(\omega) + \mathcal{L}_l^o\{f_3e_1e_2\}(\omega)\mathcal{L}_l^o\{g\}(\omega) + \mathcal{L}_l^o\{f_4e_4\}(\omega)\mathcal{L}_l^o\{g\}(\omega) \\ & + \mathcal{L}_l^o\{f_5e_1e_4\}(\omega)\mathcal{L}_l^o\{g\}(\omega) + \mathcal{L}_l^o\{f_6e_2e_4\}(\omega)\mathcal{L}_l^o\{g\}(\omega) + \mathcal{L}_l^o\{f_7e_1e_2e_4\}(\omega)\mathcal{L}_l^o\{g\}(\omega)). \end{aligned}$$

This concludes the proof.

5. Applications of 3-D linear time-invariant systems

In this section, on the basis of the properties, and the new convolution operation for the LOCLCT, we discuss the 3D LTI in the $\mathbb{F}$ domain. The results can simplify the representation of 3-D LTI systems.

Theorem 6. Let $f\Theta g = h$, $\mathcal{L}_l^o\{h\}(\omega)$ denote the LOCLCT of $f\Theta g$. Then,

$$\mathcal{H}(\omega_1, \omega_2, \omega_3) = [\mathcal{G}(\omega_1, \omega_2, \omega_3) \cdot (\mathcal{Z}_{leee}(\omega) + \mathcal{Z}_{leeo}(\omega)e_4) + \mathcal{G}(\omega_1, \omega_2, -\omega_3) \cdot (\mathcal{Z}_{leoe}(\omega)e_2 + \mathcal{Z}_{leoo}(\omega)e_6)]$$

$$+ \mathcal{G}(\omega_1, -\omega_2, -\omega_3) \cdot (\mathcal{Z}_{loee}(\omega)e_1 + \mathcal{Z}_{looe}(\omega)e_3 + \mathcal{Z}_{loeo}(\omega)e_5 + \mathcal{Z}_{looo}(\omega)e_7)],$$

where $\mathcal{H}(\omega) = \mathcal{L}_f^0\{h\}(\omega)$, $\mathcal{G}(\omega) = \mathcal{L}_f^0\{g\}(\omega)$, $\mathcal{Z}(\omega) = \mathcal{L}_f^0\{f\}(\omega) = \mathcal{Z}_{leee}(\omega) + \mathcal{Z}_{leeo}(\omega) + \mathcal{Z}_{leoe}(\omega) + \mathcal{Z}_{leoo}(\omega) + \mathcal{Z}_{loee}(\omega) + \mathcal{Z}_{loeo}(\omega) + \mathcal{Z}_{looe}(\omega) + \mathcal{Z}_{looo}(\omega)$.

Proof. In the subsequent proof process, the facts listed below are employed, for every $m_1, m_2, m_3 \in \mathbb{R}$, we have

$$((e^{e_1 m_1}) \cdot (e^{e_2 m_2})) \cdot (e^{e_4 m_3}) = ((e^{e_1 m_1} \cdot e^{e_2 m_2}) \cdot e^{e_4 m_3}), \quad (40)$$

$$((e^{e_1 m_1} \cdot e_1) \cdot (e^{e_2 m_2})) \cdot (e^{e_4 m_3}) = ((e^{e_1 m_1} \cdot e^{-e_2 m_2}) \cdot e^{-e_4 m_3}) \cdot e_1, \quad (41)$$

$$((e^{e_1 m_1}) \cdot (e^{e_2 m_2} \cdot e_2)) \cdot (e^{e_4 m_3}) = ((e^{e_1 m_1} \cdot e^{e_2 m_2}) \cdot e^{-e_4 m_3}) \cdot e_2, \quad (42)$$

$$((e^{e_1 m_1} \cdot e_1) \cdot (e^{e_2 m_2} \cdot e_2)) \cdot (e^{e_4 m_3}) = ((e^{e_1 m_1} \cdot e^{-e_2 m_2}) \cdot e^{-e_4 m_3}) \cdot e_3, \quad (43)$$

$$((e^{e_1 m_1}) \cdot (e^{e_2 m_2})) \cdot (e^{e_4 m_3} \cdot e_4) = ((e^{e_1 m_1} \cdot e^{e_2 m_2}) \cdot e^{e_4 m_3}) \cdot e_4, \quad (44)$$

$$((e^{e_1 m_1} \cdot e_1) \cdot (e^{e_2 m_2})) \cdot (e^{e_4 m_3} \cdot e_4) = ((e^{e_1 m_1} \cdot e^{-e_2 m_2}) \cdot e^{-e_4 m_3}) \cdot e_5, \quad (45)$$

$$((e^{e_1 m_1}) \cdot (e^{e_2 m_2} \cdot e_2)) \cdot (e^{e_4 m_3} \cdot e_4) = ((e^{e_1 m_1} \cdot e^{e_2 m_2}) \cdot e^{e_4 m_3}) \cdot e_6, \quad (46)$$

$$((e^{e_1 a_1} \cdot e_1) \cdot (e^{e_2 m_2} \cdot e_2)) \cdot (e^{e_4 m_3} \cdot e_4) = ((e^{e_1 m_1} \cdot e^{-e_2 m_2}) \cdot e^{-e_4 m_3}) \cdot e_7. \quad (47)$$

By using Eqs. (40)–(47), we can represent Theorem 5 as follows:

$$\begin{aligned} \mathcal{L}_f^0\{h\}(\omega) &= \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} K f(\sigma) g(\mathbf{x} - \sigma) \frac{1}{\sqrt{2\pi b_1}} e^{-e_1(\sigma_1(x_1 - \sigma_1)) \frac{a_1}{b_1}} \frac{1}{\sqrt{2\pi b_2}} e^{-e_2(\sigma_2(x_2 - \sigma_2)) \frac{a_2}{b_2}} \frac{1}{\sqrt{2\pi b_3}} e^{-e_4(\sigma_3(x_3 - \sigma_3)) \frac{a_3}{b_3}} d\sigma dx. \end{aligned}$$

Similarly, let $s_1 = x_1 - \sigma_1, s_2 = x_2 - \sigma_2, s_3 = x_3 - \sigma_3$, we obtain

$$\begin{aligned} \mathcal{L}_f^0\{h\}(\omega) &= D_f \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} e^{e_1 \left(\frac{a_1((s_1 + \sigma_1)^2)}{2b_1} - \frac{(s_1 + \sigma_1)\omega_1}{b_1} + \frac{d_1\omega_1^2}{2b_1} \right)} \cdot e^{e_2 \left(\frac{a_2((s_2 + \sigma_2)^2)}{2b_2} - \frac{(s_2 + \sigma_2)\omega_2}{b_2} + \frac{d_2\omega_2^2}{2b_2} \right)} \\ &\quad \cdot e^{e_4 \left(\frac{a_3((s_3 + \sigma_3)^2)}{2b_3} - \frac{(s_3 + \sigma_3)\omega_3}{b_3} + \frac{d_3\omega_3^2}{2b_3} \right)} \cdot f(\sigma) g(s) e^{-e_1(\sigma_1 s_1) \frac{a_1}{b_1}} e^{-e_2(\sigma_2 s_2) \frac{a_2}{b_2}} e^{-e_4(\sigma_3 s_3) \frac{a_3}{b_3}} d\sigma ds \\ &= e^{-\frac{e_1 d_1 \omega_1^2}{2b_1}} e^{-\frac{e_2 d_2 \omega_2^2}{2b_2}} e^{-\frac{e_4 d_3 \omega_3^2}{2b_3}} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} e^{e_1 \left(\frac{a_1 s_1^2}{2b_1} - \frac{s_1 \omega_1}{b_1} + \frac{d_1 \omega_1^2}{2b_1} \right)} e^{e_2 \left(\frac{a_2 s_2^2}{2b_2} - \frac{s_2 \omega_2}{b_2} + \frac{d_2 \omega_2^2}{2b_2} \right)} e^{e_4 \left(\frac{a_3 s_3^2}{2b_3} - \frac{s_3 \omega_3}{b_3} + \frac{d_3 \omega_3^2}{2b_3} \right)} \\ &\quad \cdot e^{e_1 \left(\frac{a_1 \sigma_1^2}{2b_1} - \frac{\sigma_1 \omega_1}{b_1} + \frac{d_1 \omega_1^2}{2b_1} \right)} e^{e_2 \left(\frac{a_2 \sigma_2^2}{2b_2} - \frac{\sigma_2 \omega_2}{b_2} + \frac{d_2 \omega_2^2}{2b_2} \right)} e^{e_4 \left(\frac{a_3 \sigma_3^2}{2b_3} - \frac{\sigma_3 \omega_3}{b_3} + \frac{d_3 \omega_3^2}{2b_3} \right)} \cdot f(\sigma) g(s) d\sigma ds. \end{aligned}$$

Let $r_k = \frac{a_k s_k^2}{2b_k} - \frac{s_k \omega_k}{b_k} + \frac{d_k \omega_k^2}{2b_k}, t_k = \frac{a_k \sigma_k^2}{2b_k} - \frac{\sigma_k \omega_k}{b_k} + \frac{d_k \omega_k^2}{2b_k}, k = 1, 2, 3$. We have

$$\begin{aligned} &= ((e^{e_1 r_1}) \cdot \cos(t_1)) \cdot ((e^{e_2 r_2}) \cdot \cos(t_2)) \cdot ((e^{e_4 r_3}) \cdot \cos(t_3)) \\ &\quad + ((e^{e_1 r_1}) \cdot e_1 \sin(t_1)) \cdot ((e^{e_2 r_2}) \cdot \cos(t_2)) \cdot ((e^{e_4 r_3}) \cdot \cos(t_3)) \\ &\quad + ((e^{e_1 r_1}) \cdot \cos(t_1)) \cdot ((e^{e_2 r_2}) \cdot e_2 \sin(t_2)) \cdot ((e^{e_4 r_3}) \cdot \cos(t_3)) \\ &\quad + ((e^{e_1 r_1}) \cdot e_1 \sin(t_1)) \cdot ((e^{e_2 r_2}) \cdot e_2 \sin(t_2)) \cdot ((e^{e_4 r_3}) \cdot \cos(t_3)) \\ &\quad + ((e^{e_1 r_1}) \cdot \cos(t_1)) \cdot ((e^{e_2 r_2}) \cdot \cos(t_2)) \cdot ((e^{e_4 r_3}) \cdot e_4 \sin(t_3)) \\ &\quad + ((e^{e_1 r_1}) \cdot e_1 \sin(t_1)) \cdot ((e^{e_2 r_2}) \cdot \cos(t_2)) \cdot ((e^{e_4 r_3}) \cdot e_4 \sin(t_3)) \\ &\quad + ((e^{e_1 r_1}) \cdot \cos(t_1)) \cdot ((e^{e_2 r_2}) \cdot e_2 \sin(t_2)) \cdot ((e^{e_4 r_3}) \cdot e_4 \sin(t_3)) \\ &\quad + ((e^{e_1 r_1}) \cdot e_1 \sin(t_1)) \cdot ((e^{e_2 r_2}) \cdot e_2 \sin(t_2)) \cdot ((e^{e_4 r_3}) \cdot e_4 \sin(t_3)). \end{aligned}$$

Using Eqs. (40)–(47), we obtain

$$\begin{aligned} \mathcal{H}(\omega_1, \omega_2, \omega_3) &= \mathcal{G}(\omega_1, \omega_2, \omega_3) \cdot (\mathcal{Z}_{leee}(\omega) + \mathcal{Z}_{leeo}(\omega)e_4) \\ &\quad + \mathcal{G}(\omega_1, \omega_2, -\omega_3) \cdot (\mathcal{Z}_{leoe}(\omega)e_2 + \mathcal{Z}_{leoo}(\omega)e_6) \\ &\quad + \mathcal{G}(\omega_1, -\omega_2, -\omega_3) \cdot (\mathcal{Z}_{loee}(\omega)e_1 + \mathcal{Z}_{looe}(\omega)e_3 + \mathcal{Z}_{loeo}(\omega)e_5 + \mathcal{Z}_{looo}(\omega)e_7). \end{aligned} \quad (48)$$

Corollary 3. If we see $\mathcal{Z}(\omega)$ and $\mathcal{G}(\omega)$ as the frequency responses for 3-D LTI systems, then we use the notation $\otimes$ to represent multiplication in the $\mathbb{F}$ field and rewrite Eq. (48) using $\otimes$ as follows:

$$\mathcal{H}(\omega) = \mathcal{Z}(\omega) \otimes \mathcal{G}(\omega).$$

This is similar to the input–output relationship of an octonion-valued signal. The input signal $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ and the output $h : \mathbb{R}^3 \rightarrow \mathbb{R}$ of the 3-D LTI systems are shown as follows:

$$h(\mathbf{x}) = \int_{\mathbb{R}^3} f(\mathbf{y}) \cdot g(\mathbf{x} - \mathbf{y}) d\mathbf{y} = (f * g)(\mathbf{x}).$$

The classical convolution operation derives the output signal from the input and impulse response; this formula is shown in Fig. 1.

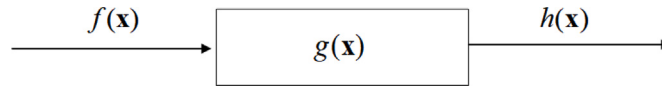


Fig. 1. 3-D LTI systems.

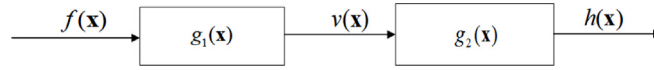


Fig. 2. Cascade connection of the LTI systems.

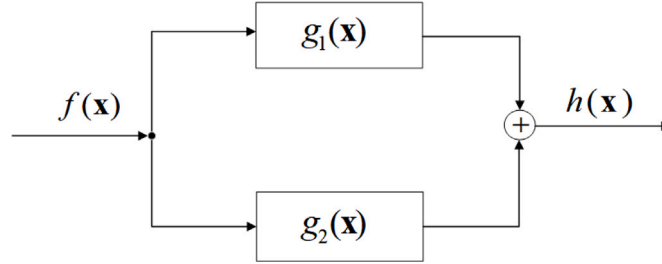


Fig. 3. Parallel connection of LTI systems.

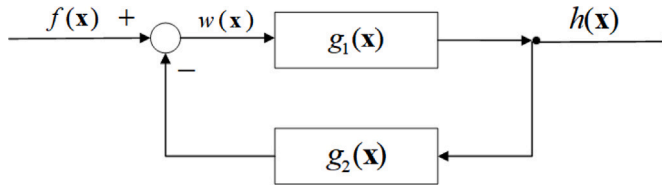


Fig. 4. Feedback connection of LTI systems.

In classical signal and system theory, there are cascade connections, parallel connections, and feedback connections between systems. Next, we discuss the mathematical forms and graphs of these three connections. The expression for a cascade connection is as follows and is shown in Fig. 2:

$$H(\omega) = \mathcal{G}_2(\omega) \otimes \mathcal{V}(\omega),$$

$$\mathcal{V}(\omega) = \mathcal{G}_1(\omega) \otimes \mathcal{Z}(\omega),$$

where

$$\mathcal{G}_2(\omega) = e^{-e_1 \frac{d_1}{2b_1} \omega_1^2} e^{-e_2 \frac{d_2}{2b_2} \omega_2^2} e^{-e_4 \frac{d_3}{2b_3} \omega_3^2} \mathcal{L}_I^o \{g_2\}(\omega),$$

$$\mathcal{V}(\omega) = e^{-e_1 \frac{d_1}{2b_1} \omega_1^2} e^{-e_2 \frac{d_2}{2b_2} \omega_2^2} e^{-e_4 \frac{d_3}{2b_3} \omega_3^2} \mathcal{L}_I^o \{v\}(\omega),$$

$$\mathcal{G}_1(\omega) = e^{-e_1 \frac{d_1}{2b_1} \omega_1^2} e^{-e_2 \frac{d_2}{2b_2} \omega_2^2} e^{-e_4 \frac{d_3}{2b_3} \omega_3^2} \mathcal{L}_I^o \{g_1\}(\omega).$$

Since the algebra of $\mathbb{F}$ is associative and commutative, we have

$$H(\omega) = \mathcal{Z}(\omega) \otimes \mathcal{G}(\omega),$$

where

$$\mathcal{G}(\omega) = \mathcal{G}_1(\omega) \otimes \mathcal{G}_2(\omega).$$

The expression for a parallel connection is as follows, and is shown in Fig. 3:

$$H(\omega) = \mathcal{Z}(\omega) \otimes \mathcal{G}(\omega),$$

where

$$\mathcal{G}(\omega) = \mathcal{G}_1(\omega) + \mathcal{G}_2(\omega).$$

Finally, we discuss the feedback connections between systems, as shown in Fig. 4. The equations of the system are expressed as follows:

$$H(\omega) = \mathcal{G}_1(\omega) \otimes \mathcal{W}(\omega), \quad \mathcal{W}(\omega) = \mathcal{Z}(\omega) - \mathcal{G}_2(\omega) \otimes H(\omega),$$

where $\mathcal{W}(\omega) = e^{-e_1 \frac{d_1}{2b_1} \omega_1^2} e^{-e_2 \frac{d_2}{2b_2} \omega_2^2} e^{-e_4 \frac{d_3}{2b_3} \omega_3^2} \mathcal{L}_I^o \{w\}(\omega).$

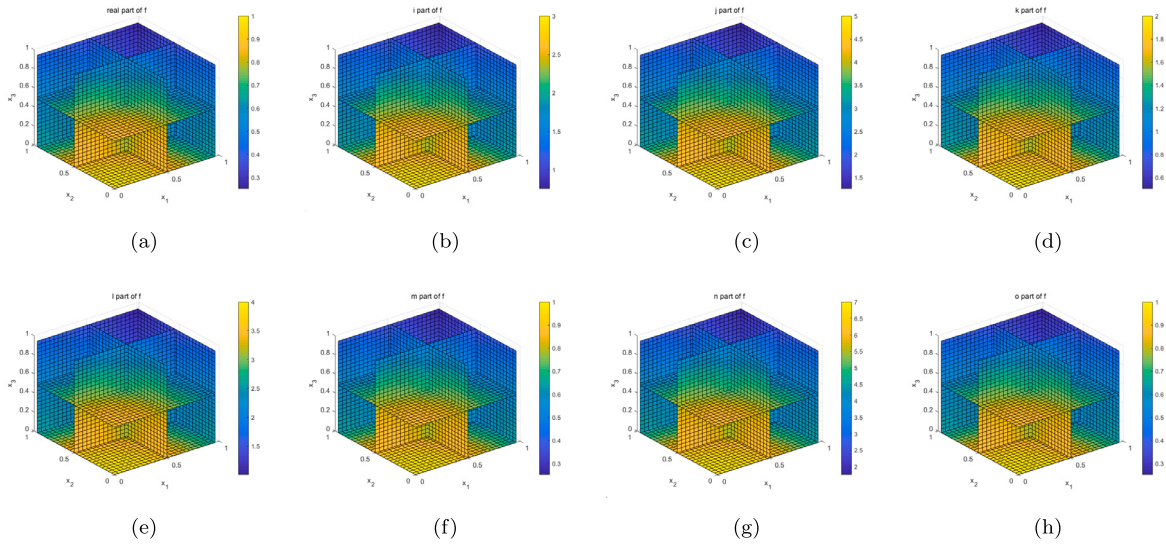


Fig. 5. (a) Real part of $f(x)$; (b) i part of $f(x)$; (c) j part of $f(x)$; (d) k part of $f(x)$; (e) l part of $f(x)$; (f) m part of $f(x)$; (g) n part of $f(x)$; (h) o part of $f(x)$.

6. Examples and simulations

In this section, the proposed LOCLCT is demonstrated through two illustrative examples.

Example 1. We consider the following octonion-valued signal:

$$f(x) = [1 + 3e_1 + 5e_2 + 2e_3 + 4e_4 + e_5 + 7e_6 + e_7]e^{\left[-\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right]} \quad (49)$$

where $\alpha, \beta, \gamma \in \mathbb{R}^+$. We plot the components of the octonion-valued signal $f(x)$, as shown in Fig. 5(a)–(h).

Next, we calculate the LOCLCT of $f(x)$ as follows:

$$\begin{aligned} \mathcal{L}_l^o\{f\}(\omega) &= \int_{\mathbb{R}^3} K f(x) dx \\ &= D_\gamma \int_{\mathbb{R}^3} e^{\left[e_1\left(\frac{a_1 x_1^2}{2b_1} - \frac{x_1 \omega_1}{b_1} + \frac{d_1}{2b_1} \omega_1^2\right)\right]} e^{\left[e_2\left(\frac{a_2 x_2^2}{2b_2} - \frac{x_2 \omega_2}{b_2} + \frac{d_2}{2b_2} \omega_2^2\right)\right]} \\ &\quad \cdot e^{\left[e_4\left(\frac{a_3 x_3^2}{2b_3} - \frac{x_3 \omega_3}{b_3} + \frac{d_3}{2b_3} \omega_3^2\right)\right]} f(x) e^{\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right]} dx \\ &= D_\gamma f(x) \int_{\mathbb{R}^3} e^{\left[-e_1\left(\frac{a_1 x_1^2}{2b_1} - \frac{x_1 \omega_1}{b_1} + \frac{d_1}{2b_1} \omega_1^2\right)\right]} e^{\left[-e_2\left(\frac{a_2 x_2^2}{2b_2} - \frac{x_2 \omega_2}{b_2} + \frac{d_2}{2b_2} \omega_2^2\right)\right]} \\ &\quad \cdot e^{\left[-e_4\left(\frac{a_3 x_3^2}{2b_3} - \frac{x_3 \omega_3}{b_3} + \frac{d_3}{2b_3} \omega_3^2\right)\right]} e^{\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right]} dx \\ &= D_\gamma f(x) \cdot \int_{\mathbb{R}^3} e^{\left[-x_1^2\left(\frac{e_1 a_1}{2b_1} + \frac{\alpha}{2}\right) - \frac{e_1 x_1 \omega_1}{b_1} + \frac{e_1 d_1 \omega_1^2}{2b_1}\right]} \\ &\quad \cdot e^{\left[-x_2^2\left(\frac{e_2 a_2}{2b_2} + \frac{\beta}{2}\right) + \frac{e_2 x_2 \omega_2}{b_2} - \frac{e_2 d_2 \omega_2^2}{2b_2}\right]} \cdot e^{\left[-x_3^2\left(\frac{e_4 a_3}{2b_3} + \frac{\gamma}{2}\right) - \frac{e_4 x_3 \omega_3}{b_3} + \frac{e_4 d_3 \omega_3^2}{2b_3}\right]} dx \\ &= f(x) e^{\left(\frac{e_1 d_1}{2b_1} \omega_1^2\right)} e^{\left(-\frac{e_2 d_2}{2b_2} \omega_2^2\right)} e^{\left(\frac{e_4 d_3}{2b_3} \omega_3^2\right)} e^{\left[-\frac{\omega_1^2}{2b_1(-e_1 a_1 + ab_1)}\right]} \frac{1}{\sqrt{-e_1 a_1 + ab_1}} \\ &\quad e^{\left[-\frac{\omega_2^2}{2b_2(-e_2 a_2 + \beta b_2)}\right]} \frac{1}{\sqrt{-e_2 a_2 + \beta b_2}} e^{\left[-\frac{\omega_3^2}{2b_3(-e_4 a_3 + \gamma b_3)}\right]} \frac{1}{\sqrt{-e_4 a_3 + \gamma b_3}} \\ &= f(x) \frac{1}{\sqrt{-e_1 a_1 + ab_1} \sqrt{-e_2 a_2 + \beta b_2} \sqrt{-e_4 a_3 + \gamma b_3}} e^{\left[-\frac{\omega_1^2}{2b_1}\left(e_1 d_1 + \frac{1}{-e_1 a_1 + ab_1}\right)\right]} \\ &\quad e^{\left[-\frac{\omega_2^2}{2b_2}\left(e_2 d_2 + \frac{1}{-e_2 a_2 + \beta b_2}\right)\right]} e^{\left[-\frac{\omega_3^2}{2b_3}\left(-e_4 d_3 - \frac{1}{-e_4 a_3 + \gamma b_3}\right)\right]}. \end{aligned}$$

Let $A_\alpha = \frac{1}{-e_1 a_1 + ab_1}$, $B_\beta = \frac{1}{-e_2 a_2 + \beta b_2}$, $C_\gamma = \frac{1}{-e_4 a_3 + \gamma b_3}$.
We have

$$\mathcal{L}_l^o\{f\}(\omega) = f(x) \sqrt{A_\alpha B_\beta C_\gamma} e^{\left[-\frac{\omega_1^2}{2b_1}(e_1 d_1 + A_\alpha)\right]} e^{\left[-\frac{\omega_2^2}{2b_2}(e_2 d_2 + B_\beta)\right]} e^{\left[-\frac{\omega_3^2}{2b_3}(e_4 d_3 - C_\gamma)\right]}.$$

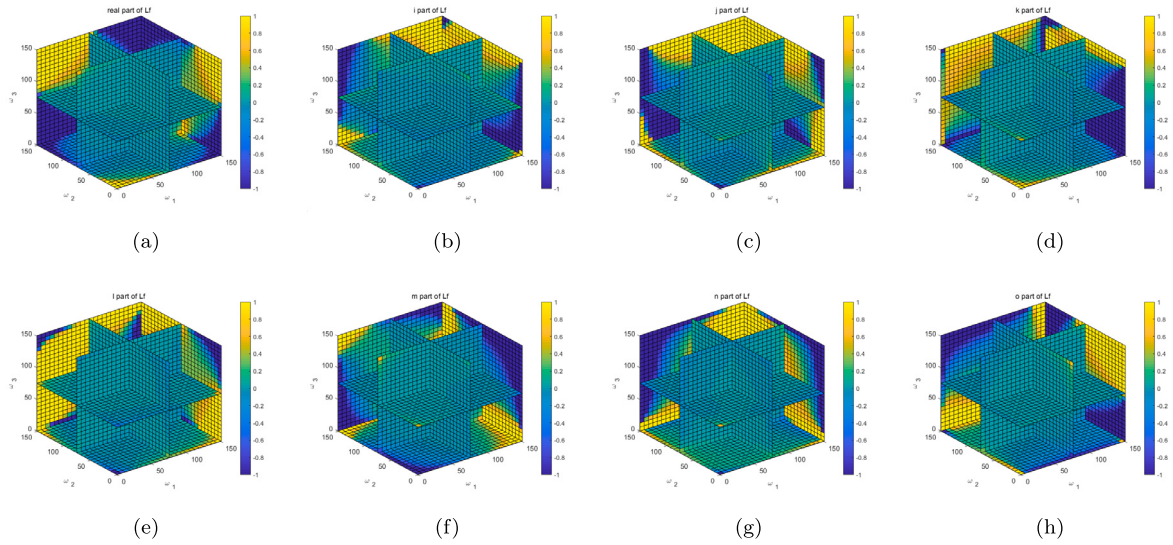


Fig. 6. (a) Real part of $\mathcal{L}_f^o(\omega)$; (b) i part of $\mathcal{L}_f^o(\omega)$; (c) j part of $\mathcal{L}_f^o(\omega)$; (d) k part of $\mathcal{L}_f^o(\omega)$; (e) l part of $\mathcal{L}_f^o(\omega)$; (f) m part of $\mathcal{L}_f^o(\omega)$; (g) n part of $\mathcal{L}_f^o(\omega)$; (h) o part of $\mathcal{L}_f^o(\omega)$.

Let $a_k = 1, b_k = 1, k_i = -1, d_k = 0, k = 1, 2, 3, \alpha = \beta = \gamma = 1$, therefore, we have:

$$\mathcal{L}_f^o(\omega) = (1 + 3e_1 + 5e_2 + 2e_3 + 4e_4 + e_5 + 7e_6 + e_7)e^{(-\frac{\omega_1^2}{4} - \frac{\omega_2^2}{4} + \frac{\omega_3^2}{4})} \cdot e^{(e_1 \frac{\omega_1^2}{4})} e^{(-e_2 \frac{\omega_2^2}{4})} e^{(-e_4 \frac{\omega_3^2}{4})}.$$

Setting $\eta_1 = -\frac{\omega_1^2}{4}, \eta_2 = -\frac{\omega_2^2}{4}, \eta_3 = \frac{\omega_3^2}{4}$, we have

$$\begin{aligned} \mathcal{L}_f^o(\omega) &= (1 + 3e_1 + 5e_2 + 2e_3 + 4e_4 + e_5 + 7e_6 + e_7) \sqrt{\frac{(1 + e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + e_7)}{8}} \\ &\quad \cdot e^{\eta_1 + \eta_2 + \eta_3} (\cos \eta_1 - e_1 \sin \eta_1) (\cos \eta_2 + e_2 \sin \eta_2) (\cos \eta_3 - e_4 \sin \eta_3) \\ &= E_0(\eta) (ccc + 4ccs - 5csc + 7css - scs + 3scc + 2ssc - sss) \\ &\quad + E_0(\eta) e_1 (3ccc + ccs - 2csc - css + 4scs - scc - 5ssc - 7sss) \\ &\quad + E_0(\eta) e_2 (5ccc + 7ccs + csc - 4css - scs - 2scc + 3ssc + 5sss) \\ &\quad + E_0(\eta) e_3 (2ccc + ccs + 3csc + css + 7scs + 5scc - ssc + 4sss) \\ &\quad + E_0(\eta) e_4 (4ccc - ccs + 7csc + 5css - 3scs - scc - ssc - 2sss) \\ &\quad + E_0(\eta) e_5 (ccc - 3ccs + csc - 2css + scs + 4scc + 7ssc - 5sss) \\ &\quad + E_0(\eta) e_6 (7ccc - 5ccs - 4csc - css - 2scs + scc - ssc + 3sss) \\ &\quad + E_0(\eta) e_7 (ccc - 2ccs - csc + 3css + 5scs - 7scc + 4ssc + sss), \end{aligned}$$

where

$$E(\eta) = \sqrt{\frac{(1 + e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + e_7)}{8}} e^{\eta_1 + \eta_2 + \eta_3}.$$

Remark 4. In the above computation process, we define $s_{(\eta_1)\eta_2\eta_3} = \sin \eta_1 \eta_2 \eta_3, c_{(\eta_1)\eta_2\eta_3} = \cos \eta_1 \eta_2 \eta_3, y_{(\eta_1)\eta_2\eta_3} \in \{c, s\}, i = 2, 3$. The c, s of the above formula denotes the parity function, such as $scc = \sin \eta_1 \cos \eta_2 \sin \eta_3, csc = \cos \eta_1 \sin \eta_2 \cos \eta_3$.

We plot the frequency spectrum of the LOCLCT, as shown in Fig. 6(a)–(h). For a convenient comparison, we also plot the frequency spectrum of the LOCFT in Fig. 7(a)–(h).

Example 2. We consider the octonion-valued signal as follows:

$$\begin{aligned} f(\mathbf{x}) &= [1 + x_1 e_1 + x_2 e_2 + x_3 e_3 + (1 + x_1)^2 e_4 + (1 + x_2)^2 e_5 + (1 + x_3)^2 e_6 + (1 + x_1 x_2 x_3) e_7] \\ &\quad \cdot e^{[-(\frac{a}{2} x_1^2 + \frac{b}{2} x_2^2 + \frac{c}{2} x_3^2)]}. \end{aligned} \quad (50)$$

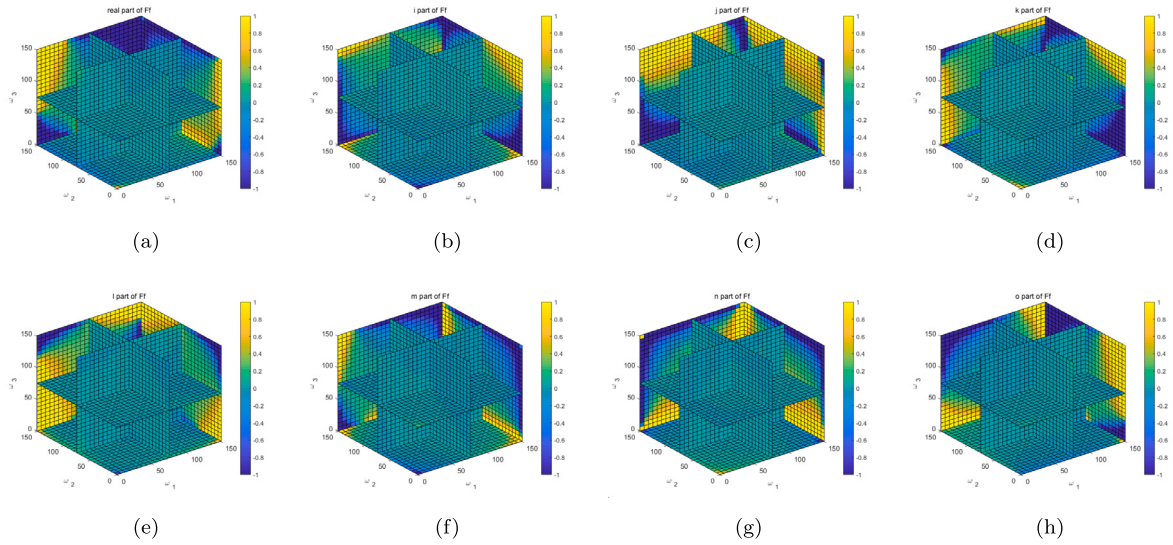


Fig. 7. (a) Real part of $F_l^o\{f\}(\omega)$; (b) i part of $F_l^o\{f\}(\omega)$; (c) j part of $F_l^o\{f\}(\omega)$; (d) k part of $F_l^o\{f\}(\omega)$; (e) l part of $F_l^o\{f\}(\omega)$; (f) m part of $F_l^o\{f\}(\omega)$; (g) n part of $F_l^o\{f\}(\omega)$; (h) o part of $F_l^o\{f\}(\omega)$.

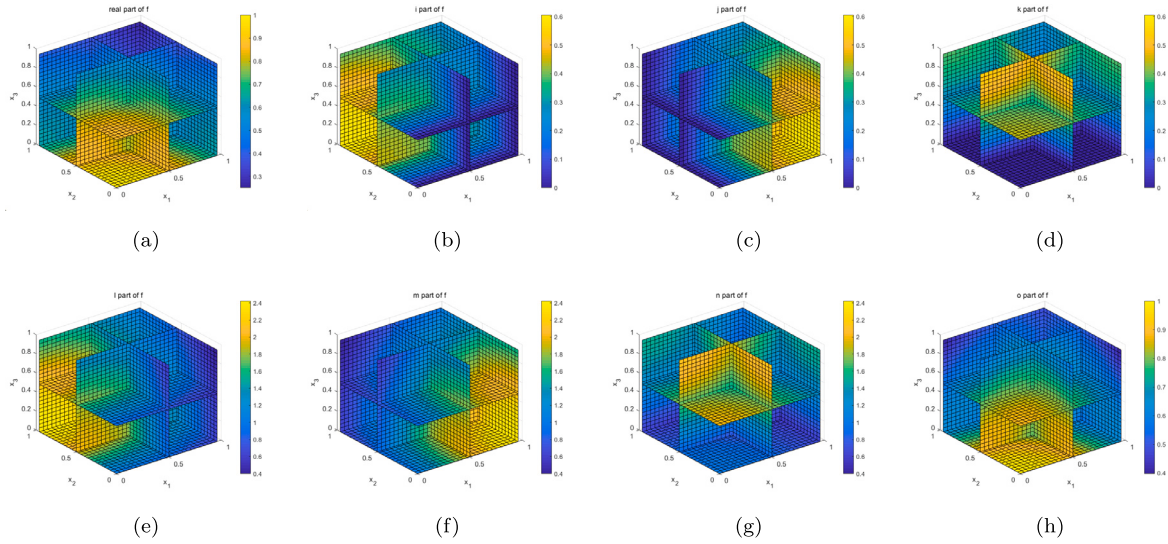


Fig. 8. (a) Real part of $f(x)$; (b) i part of $f(x)$; (c) j part of $f(x)$; (d) k part of $f(x)$; (e) l part of $f(x)$; (f) m part of $f(x)$; (g) n part of $f(x)$; (h) o part of $f(x)$.

The spectrum of $f(x)$ is shown in Fig. 8(a)–(h). The LOCLCT of $f(x)$ is derived as follows:

$$\begin{aligned}
 \mathcal{L}_l^o\{f\}(\omega) &= \int_{\mathbb{R}^3} K[1 + x_1 e_1 + x_2 e_2 + x_3 e_3 + (1 + x_1)^2 e_4 + (1 + x_2)^2 e_5 \\
 &\quad + (1 + x_3)^2 e_6 + (1 + x_1 x_2 x_3) e_7] \cdot e\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right] dx \\
 &= \int_{\mathbb{R}^3} Ke\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right] \cdot [1 - x_1 e_1 - x_2 e_2 - x_3 e_3 - (1 + x_1)^2 e_4 \\
 &\quad - (1 + x_2)^2 e_5 - (1 + x_3)^2 e_6 - (1 + x_1 x_2 x_3) e_7] dx \\
 &= \int_{\mathbb{R}^3} Ke\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right] dx - \int_{\mathbb{R}^3} Ke\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right] x_1 e_1 dx - \int_{\mathbb{R}^3} Ke\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right] x_2 e_2 dx \\
 &\quad - \int_{\mathbb{R}^3} Ke\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right] x_3 e_3 dx - \int_{\mathbb{R}^3} Ke\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right] (1 + x_1)^2 e_4 dx \\
 &\quad - \int_{\mathbb{R}^3} Ke\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right] (1 + x_2)^2 e_5 dx - \int_{\mathbb{R}^3} Ke\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right] (1 + x_3)^2 e_6 dx \\
 &\quad - \int_{\mathbb{R}^3} Ke\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right] (1 + x_1 x_2 x_3) e_7 dx.
 \end{aligned}$$

We denote the above eight integrals as $I_0, I_1, I_2, I_3, I_4, I_5, I_6, I_7$. Therefore, $\mathcal{L}_l^o\{f\}(\omega) = I_0 + I_1 + I_2 + I_3 + I_4 + I_5 + I_6 + I_7$. The results of the above integral can be represented by

$$\begin{aligned}
 I_0 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} d\mathbf{x} \\
 &= \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} \int_{\mathbb{R}^3} e^{\left[e_1\left(\frac{a_1x_1^2}{2b_1} - \frac{x_1\omega_1}{b_1} + \frac{d_1}{2b_1}\omega_1^2\right)\right]} e^{\left[e_2\left(\frac{a_2x_2^2}{2b_2} - \frac{x_2\omega_2}{b_2} + \frac{d_2}{2b_2}\omega_2^2\right)\right]} e^{\left[e_4\left(\frac{a_3x_3^2}{2b_3} - \frac{x_3\omega_3}{b_3} + \frac{d_3}{2b_3}\omega_3^2\right)\right]} \cdot e^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} d\mathbf{x} \\
 &= \frac{1}{\sqrt{e_1a_1 + \alpha b_1}\sqrt{-e_2a_2 + \beta b_2}\sqrt{e_4a_3 + \gamma b_3}} e^{\left[\frac{\omega_1^2}{2b_1}\left(e_1d_1 - \frac{1}{e_1a_1 + \alpha b_1}\right)\right]} e^{\left[\frac{\omega_2^2}{2b_2}\left(-e_2d_2 + \frac{1}{-e_2a_2 + \beta b_2}\right)\right]} \\
 &\quad e^{\left[\frac{\omega_3^2}{2b_3}\left(e_4d_3 - \frac{1}{e_4a_3 + \gamma b_3}\right)\right]}, \\
 I_1 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} x_1 e_1 d\mathbf{x} = 0, \\
 I_2 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} x_2 e_2 d\mathbf{x} = 0, \\
 I_3 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} x_3 e_3 d\mathbf{x} = 0, \\
 I_4 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} (1+x_1)^2 e_4 d\mathbf{x} \\
 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} e_4 d\mathbf{x} + \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} 2x_1 e_4 d\mathbf{x} + \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} x_1^2 e_4 d\mathbf{x} \\
 &= \frac{1}{\sqrt{e_1a_1 + \alpha b_1}\sqrt{-e_2a_2 + \beta b_2}\sqrt{e_4a_3 + \gamma b_3}} e^{\left[\frac{\omega_1^2}{2b_1}\left(e_1d_1 - \frac{1}{e_1a_1 + \alpha b_1}\right)\right]} e^{\left[\frac{\omega_2^2}{2b_2}\left(-e_2d_2 + \frac{1}{-e_2a_2 + \beta b_2}\right)\right]} \\
 &\quad \cdot e^{\left[\frac{\omega_3^2}{2b_3}\left(e_4d_3 - \frac{1}{e_4a_3 + \gamma b_3}\right)\right]} e_4 + D_\gamma e^{\left(-e_1\frac{d_1}{2b_1}\omega_1^2\right)} e^{\left(e_2\frac{d_2}{2b_2}\omega_2^2\right)} e^{\left(-e_4\frac{d_3}{2b_3}\omega_3^2\right)} \\
 &\quad \cdot \left[\sqrt{\frac{2b_1}{-e_1a_1 + \alpha b_1}} \cdot e^{\left(\frac{\omega_1^2}{2b_1(-e_1a_1 + \alpha b_1)}\right)} \cdot \left(\frac{\pi b_1}{-e_1a_1 + \alpha b_1} \sqrt{2b_1\left(\frac{e_1\omega_1^2}{-e_1a_1 + \alpha b_1}\right)^3} + \sqrt{\frac{2\pi b_1\omega_1^4}{(-e_1a_1 + \alpha b_1)^5}} \right) \right] e_4, \\
 I_5 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} (1+x_2)^2 e_5 d\mathbf{x} + D_\gamma e^{\left(-e_1\frac{d_1}{2b_1}\omega_1^2\right)} e^{\left(e_2\frac{d_2}{2b_2}\omega_2^2\right)} e^{\left(-e_4\frac{d_3}{2b_3}\omega_3^2\right)} \\
 &\quad \cdot \left[\sqrt{\frac{2b_2}{-e_2a_2 + \beta b_2}} \cdot e^{\left(\frac{\omega_2^2}{2b_2(-e_2a_2 + \beta b_2)}\right)} \cdot \left(\frac{\pi b_2}{-e_2a_2 + \beta b_2} \sqrt{2b_2\left(\frac{e_2\omega_2^2}{-e_2a_2 + \beta b_2}\right)^3} + \sqrt{\frac{2\pi b_2\omega_2^4}{(-e_2a_2 + \beta b_2)^5}} \right) \right] e_5 \\
 &= \frac{1}{\sqrt{e_1a_1 + \alpha b_1}\sqrt{-e_2a_2 + \beta b_2}\sqrt{e_4a_3 + \gamma b_3}} e^{\left[\frac{\omega_1^2}{2b_1}\left(e_1d_1 - \frac{1}{e_1a_1 + \alpha b_1}\right)\right]} e^{\left[\frac{\omega_2^2}{2b_2}\left(-e_2d_2 + \frac{1}{-e_2a_2 + \beta b_2}\right)\right]} \\
 &\quad \cdot e^{\left[\frac{\omega_3^2}{2b_3}\left(e_4d_3 - \frac{1}{e_4a_3 + \gamma b_3}\right)\right]} e_5 + D_\gamma e^{\left(-e_1\frac{d_1}{2b_1}\omega_1^2\right)} e^{\left(e_2\frac{d_2}{2b_2}\omega_2^2\right)} e^{\left(-e_4\frac{d_3}{2b_3}\omega_3^2\right)} \\
 &\quad \cdot \left[\sqrt{\frac{2b_2}{-e_2a_2 + \beta b_2}} \cdot e^{\left(\frac{\omega_2^2}{2b_2(-e_2a_2 + \beta b_2)}\right)} \cdot \left(\frac{\pi b_2}{-e_2a_2 + \beta b_2} \sqrt{2b_2\left(\frac{e_2\omega_2^2}{-e_2a_2 + \beta b_2}\right)^3} + \sqrt{\frac{2\pi b_2\omega_2^4}{(-e_2a_2 + \beta b_2)^5}} \right) \right] e_5, \\
 I_6 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} (1+x_3)^2 e_6 d\mathbf{x} + \frac{1}{2\pi\sqrt{2\pi|b_1b_2b_3|}} e^{\left(-e_1\frac{d_1}{2b_1}\omega_1^2\right)} e^{\left(e_2\frac{d_2}{2b_2}\omega_2^2\right)} e^{\left(-e_4\frac{d_3}{2b_3}\omega_3^2\right)} \\
 &\quad \cdot \left[\sqrt{\frac{2b_3}{-e_4a_3 + \gamma b_3}} \cdot e^{\left(\frac{\omega_3^2}{2b_3(-e_4a_3 + \gamma b_3)}\right)} \cdot \left(\frac{\pi b_3}{-e_4a_3 + \gamma b_3} \sqrt{2b_3\left(\frac{e_4\omega_3^2}{-e_4a_3 + \gamma b_3}\right)^3} + \sqrt{\frac{2\pi b_3\omega_3^4}{(-e_4a_3 + \gamma b_3)^5}} \right) \right] e_6 \\
 &= \frac{1}{\sqrt{e_1a_1 + \alpha b_1}\sqrt{-e_2a_2 + \beta b_2}\sqrt{e_4a_3 + \gamma b_3}} e^{\left[\frac{\omega_1^2}{2b_1}\left(e_1d_1 - \frac{1}{e_1a_1 + \alpha b_1}\right)\right]} e^{\left[\frac{\omega_2^2}{2b_2}\left(-e_2d_2 + \frac{1}{-e_2a_2 + \beta b_2}\right)\right]} \\
 &\quad \cdot e^{\left[\frac{\omega_3^2}{2b_3}\left(e_4d_3 - \frac{1}{e_4a_3 + \gamma b_3}\right)\right]} e_6 + D_\gamma e^{\left(-e_1\frac{d_1}{2b_1}\omega_1^2\right)} e^{\left(e_2\frac{d_2}{2b_2}\omega_2^2\right)} e^{\left(-e_4\frac{d_3}{2b_3}\omega_3^2\right)} \\
 &\quad \cdot \left[\sqrt{\frac{2b_3}{-e_4a_3 + \gamma b_3}} \cdot e^{\left(\frac{\omega_3^2}{2b_3(-e_4a_3 + \gamma b_3)}\right)} \cdot \left(\frac{\pi b_3}{-e_4a_3 + \gamma b_3} \sqrt{2b_3\left(\frac{e_4\omega_3^2}{-e_4a_3 + \gamma b_3}\right)^3} + \sqrt{\frac{2\pi b_3\omega_3^4}{(-e_4a_3 + \gamma b_3)^5}} \right) \right] e_6, \\
 I_7 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} (1+x_1x_2x_3) e_7 d\mathbf{x}
 \end{aligned}$$

$$= \frac{1}{\sqrt{e_1 a_1 + \alpha b_1} \sqrt{-e_2 a_2 + \beta b_2} \sqrt{e_4 a_3 + \gamma b_3}} e^{\left[\frac{\omega_1^2}{2b_1} \left(e_1 d_1 - \frac{1}{e_1 a_1 + \alpha b_1}\right)\right]} \cdot e^{\left[\frac{\omega_2^2}{2b_2} \left(-e_2 d_2 + \frac{1}{-e_2 a_2 + \beta b_2}\right)\right]} e^{\left[\frac{\omega_3^2}{2b_3} \left(e_4 d_3 - \frac{1}{e_4 a_3 + \gamma b_3}\right)\right]} e_7.$$

Let $a_k = 1, b_k = 1, c_k = -1, d_k = 0, k = 1, 2, 3, \alpha = \beta = \gamma = 1, \theta_1 = \frac{\omega_1^2}{4}, \theta_2 = -\frac{\omega_2^2}{4}, \theta_3 = \frac{\omega_3^2}{4}$, we plot the LOCLCT-frequency spectrum and the LOCFT-frequency spectrum for [Example 2](#), as shown in [Figs. 9](#) and [10](#), respectively. We obtain the following computation results:

$$\begin{aligned} I_0 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right]} d\mathbf{x} \\ &= \frac{1}{\sqrt{e_1 + 1} \sqrt{-e_2 + 1} \sqrt{e_4 + 1}} e^{\left[\frac{\omega_1^2}{2} \left(-\frac{1}{e_1 + 1}\right)\right]} e^{\left[\frac{\omega_2^2}{2} \left(\frac{1}{-e_2 + 1}\right)\right]} e^{\left[\frac{\omega_3^2}{2} \left(-\frac{1}{e_4 + 1}\right)\right]} \\ &= \frac{1}{\sqrt{e_1 + 1} \sqrt{-e_2 + 1} \sqrt{e_4 + 1}} e^{-\theta_1 + \theta_2 + \theta_3} e^{-e_1 \theta_1} e^{e_2 \theta_2} e^{-e_4 \theta_3} \\ &= \sqrt{\frac{1}{(1 - e_1 + e_2 - e_3 - e_4 + e_5 - e_6 + e_7)}} (1 - e_1 + e_2 - e_3 - e_4 + e_5 - e_6 + e_7) e^{-\theta_1 + \theta_2 + \theta_3} \\ &\quad \cdot (\cos \theta_1 - e_1 \sin \theta_1) (\cos \theta_2 + e_2 \sin \theta_2) (\cos \theta_3 - e_4 \sin \theta_3) \\ &= E_0(\theta) (ccc - ccs - csc - css - scs - sec - ssc - sss) \\ &\quad + E_0(\theta) e_1 (-ccc + ccs + csc - css - scs - sec - ssc + sss) \\ &\quad + E_0(\theta) e_2 (ccc - ccs + csc + css - scs + sec - ssc + sss) \\ &\quad + E_0(\theta) e_3 (-ccc + ccs - csc + css - scs + sec - ssc - sss) \\ &\quad + E_0(\theta) e_4 (-ccc - ccs - csc + css + scs - sec - ssc + sss) \\ &\quad + E_0(\theta) e_5 (ccc + ccs + csc + css + scs - sec - ssc - sss) \\ &\quad + E_0(\theta) e_6 (-ccc - ccs + csc - css + scs + sec - ssc - sss) \\ &\quad + E_0(\theta) e_7 (ccc + ccs - csc - css + scs - sec - ssc + sss), \\ I_1 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right]} x_1 e_1 d\mathbf{x} = 0, \\ I_2 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right]} x_2 e_2 d\mathbf{x} = 0, \\ I_3 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right]} x_3 e_3 d\mathbf{x} = 0, \\ I_4 &= \int_{\mathbb{R}^3} K e^{\left[-\left(\frac{\alpha}{2} x_1^2 + \frac{\beta}{2} x_2^2 + \frac{\gamma}{2} x_3^2\right)\right]} (1 + x_1)^2 e_4 d\mathbf{x} \\ &= \frac{1}{\sqrt{e_1 + 1} \sqrt{-e_2 + 1} \sqrt{e_4 + 1}} e^{\left[\frac{\omega_1^2}{2} \left(-\frac{1}{e_1 + 1}\right)\right]} e^{\left[\frac{\omega_2^2}{2} \left(\frac{1}{-e_2 + 1}\right)\right]} e^{\left[\frac{\omega_3^2}{2} \left(-\frac{1}{e_4 + 1}\right)\right]} e_4 \\ &\quad + \frac{1}{2\pi \sqrt{2\pi}} \cdot \left[\sqrt{\frac{2}{-e_1 + 1}} \cdot e^{\left(\frac{\omega_1^2}{2(-e_1 + 1)}\right)} \cdot \left(\frac{\pi}{-e_1 + 1}\right) \sqrt{2 \left(\frac{e_1 \omega_1^2}{-e_1 + 1}\right)^3} + \sqrt{\frac{2\pi \omega_1^4}{(-e_1 + 1)^5}} \right] e_4 \\ &= \frac{e_4}{\sqrt{e_1 + 1} \sqrt{-e_2 + 1} \sqrt{e_4 + 1}} e^{\theta_1 - \theta_2 - \theta_3} e^{e_1 \theta_1} e^{-e_2 \theta_2} e^{e_4 \theta_3} \\ &\quad + \left[\frac{e_4 \sqrt{(e_1 + 1)^3}}{8 \sqrt{2\pi}} \sqrt{[(e_1 - 1) \omega_1^2]^3} + \frac{(e_1 + 1) e_4}{\sqrt{2}} \cdot \frac{\omega_1}{2\pi(-e_1 + 1)} \right] \cdot e^{-\theta_1} e^{-e_1 \theta_1} \\ &= \sqrt{\frac{1}{(-1 + e_1 + e_2 + e_3 - e_4 + e_5 - e_6 - e_7)}} (-1 + e_1 + e_2 + e_3 - e_4 + e_5 - e_6 - e_7) e^{-\theta_1 + \theta_2 + \theta_3} \\ &\quad \cdot (\cos \theta_1 + e_1 \sin \theta_1) (\cos \theta_2 - e_2 \sin \theta_2) (\cos \theta_3 + e_4 \sin \theta_3) \\ &\quad + e_5 \left(\sqrt{\frac{4}{\pi}} \theta_2^{\frac{3}{2}} + \sqrt{\frac{-\theta_2}{2\pi^2}} \right) e^{-\theta_2} (\cos \theta_2 - e_2 \sin \theta_2) \\ &= E_1(\theta) (ccc - ccs - csc - css - scs - sec + ssc + sss) \\ &\quad + E_1(\theta) e_1 (ccc - ccs + csc - css + scs + sec + ssc - sss) \\ &\quad + E_1(\theta) e_2 (-ccc + ccs - csc - css - scs + sec + ssc - sss) \\ &\quad + E_1(\theta) e_3 (ccc + ccs - csc + css - scs + sec - ssc - sss) \\ &\quad + E_1(\theta) e_4 (ccc + ccs + csc - css - scs + sec - ssc + sss) \\ &\quad + E_1(\theta) e_5 (ccc + ccs - csc - css - scs - sec + ssc - sss) \\ &\quad + E_1(\theta) e_6 (-ccc - ccs + csc - css - scs - sec - ssc - sss) \\ &\quad + E_1(\theta) e_7 (ccc + ccs + csc + css - scs - sec + ssc - sss) \\ &\quad + e_4 \sqrt{-\frac{2}{\pi}} \theta_2^{\frac{5}{4}} e^{-\theta_1} (\cos \theta_1 - e_1 \sin \theta_1), \end{aligned}$$

$$\begin{aligned}
I_5 &= \int_{\mathbb{R}^3} \text{Ke}^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} (1+x_2)^2 e_5 \mathbf{d}\mathbf{x} \\
&= \frac{1}{\sqrt{e_1+1}\sqrt{-e_2+1}\sqrt{e_4+1}} e^{\left[\frac{\omega_1^2}{2}\left(-\frac{1}{e_1+1}\right)\right]} e^{\left[\frac{\omega_2^2}{2}\left(-\frac{1}{-e_2+1}\right)\right]} e^{\left[\frac{\omega_3^2}{2}\left(-\frac{1}{e_4+1}\right)\right]} e_5 \\
&\quad + \frac{1}{2\pi\sqrt{2\pi}} \cdot \left[\sqrt{\frac{2}{-e_2+1}} \cdot e^{\left(\frac{\omega_2^2}{2(-e_2+1)}\right)} \cdot \left(\frac{\pi}{-e_2+1}\sqrt{2\left(\frac{e_2\omega_2^2}{-e_2+1}\right)^3} + \sqrt{\frac{2\pi\omega_4^4}{(-e_2+1)^5}}\right) \right] e_5 \\
&= \frac{e_5}{\sqrt{e_1+1}\sqrt{-e_2+1}\sqrt{e_4+1}} e^{\theta_1-\theta_2-\theta_3} e^{e_1\theta_1} e^{-e_2\theta_2} e^{e_4\theta_3} \\
&\quad + \left[\frac{e_5\sqrt{(e_2+1)^3}}{8\sqrt{2\pi}} \sqrt{[(e_2-1)\omega_2^2]^3} + \frac{(e_2+1)e_5}{\sqrt{2}} \cdot \frac{\omega_2}{2\pi(-e_2+1)} \right] \cdot e^{-\theta_2} e^{-e_2\theta_2} \\
&= \sqrt{\frac{1}{(-1+e_1+e_2+e_3-e_4+e_5-e_6-e_7)}} (-1+e_1+e_2+e_3-e_4+e_5-e_6-e_7) e^{-\theta_1+\theta_2+\theta_3} \\
&\quad \cdot (\cos\theta_1+e_1\sin\theta_1)(\cos\theta_2-e_2\sin\theta_2)(\cos\theta_3+e_4\sin\theta_3) + e_5 \left(\sqrt{\frac{4}{\pi}}\theta_2^{\frac{3}{2}} + \sqrt{\frac{-\theta_2}{2\pi^2}} \right) e^{-\theta_2}(\cos\theta_2-e_2\sin\theta_2) \\
&= E_2(\theta)(-ccc+ccs+csc-css-scs-scc+ssc-sss) \\
&\quad + E_2(\theta)e_1(ccc-ccs+csc+css-scs-scc-ssc-sss) \\
&\quad + E_2(\theta)e_2(ccc+ccs+csc+css+scs+scc+ssc-sss) \\
&\quad + E_2(\theta)e_3(ccc+ccs-csc+css-scs-scc+ssc+sss) \\
&\quad + E_2(\theta)e_4(-ccc-ccs+csc+css-scs+scc+ssc+sss) \\
&\quad + E_2(\theta)e_5(ccc+ccs+csc-css-scs-scc+ssc+sss) \\
&\quad + E_2(\theta)e_6(-ccc+ccs-csc+css+scs+scc+ssc+sss) \\
&\quad + E_2(\theta)e_7(-ccc+ccs+csc+css+scs-scc-ssc+sss) \\
&\quad + e_5 \left(\sqrt{\frac{4}{\pi}}\theta_2^{\frac{3}{2}} + \sqrt{\frac{-\theta_2}{2\pi^2}} \right) e^{-\theta_2}(\cos\theta_2-e_2\sin\theta_2), \\
I_6 &= \int_{\mathbb{R}^3} \text{Ke}^{\left[-\left(\frac{\alpha}{2}x_1^2 + \frac{\beta}{2}x_2^2 + \frac{\gamma}{2}x_3^2\right)\right]} (1+x_3)^2 e_6 \mathbf{d}\mathbf{x} \\
&= \frac{1}{\sqrt{e_1+1}\sqrt{-e_2+1}\sqrt{e_4+1}} e^{\left[\frac{\omega_1^2}{2}\left(-\frac{1}{e_1+1}\right)\right]} e^{\left[\frac{\omega_2^2}{2}\left(-\frac{1}{-e_2+1}\right)\right]} e^{\left[\frac{\omega_3^2}{2}\left(-\frac{1}{e_4+1}\right)\right]} e_6 \\
&\quad + \frac{1}{2\pi\sqrt{2\pi}} \cdot \left[\sqrt{\frac{2}{-e_4+1}} \cdot e^{\left(\frac{\omega_4^2}{2(-e_4+1)}\right)} \cdot \left(\frac{\pi}{-e_4+1}\sqrt{2\left(\frac{e_4\omega_3^2}{-e_4+1}\right)^3} + \sqrt{\frac{2\pi\omega_3^4}{(-e_4+1)^5}}\right) \right] e_6 \\
&= \frac{e_6}{\sqrt{e_1+1}\sqrt{-e_2+1}\sqrt{e_4+1}} e^{\theta_1-\theta_2-\theta_3} e^{e_1\theta_1} e^{-e_2\theta_2} e^{e_4\theta_3} \\
&\quad + \left[\frac{e_6\sqrt{(e_3+1)^3}}{8\sqrt{2\pi}} \sqrt{[(e_3-1)\omega_3^2]^3} + \frac{(e_4+1)e_6}{\sqrt{2}} \cdot \frac{\omega_3}{2\pi(-e_4+1)} \right] \cdot e^{-\theta_3} e^{-e_4\theta_3} \\
&= \sqrt{\frac{1}{(1-e_1+e_2+e_3+e_4+e_5+e_6+e_7)}} (1-e_1+e_2+e_3+e_4+e_5+e_6+e_7) e^{-\theta_1+\theta_2+\theta_3} \\
&\quad \cdot (\cos\theta_1+e_1\sin\theta_1)(\cos\theta_2-e_2\sin\theta_2)(\cos\theta_3+e_4\sin\theta_3) + e_6\theta_3^{\frac{1}{2}} \left(\sqrt{\frac{1}{\sqrt{2\pi}}} - 4\theta^{\frac{1}{4}}\sqrt{-\frac{2}{\pi}} \right) e^{-\theta_3}(\cos\theta_3-e_4\sin\theta_3) \\
&= E_3(\theta)(-ccc+ccs-csc+css-scs-scc+ssc+sss) \\
&\quad + E_3(\theta)e_1(-ccc-ccs+csc-css-scs-scc+ssc+sss) \\
&\quad + E_3(\theta)e_2(-ccc-ccs+csc+css-scs+scc-ssc-sss) \\
&\quad + E_3(\theta)e_3(ccc-ccs-csc+css+scs+scc+ssc+sss) \\
&\quad + E_3(\theta)e_4(-ccc-ccs-csc+css+scs+scc-ssc+sss) \\
&\quad + E_3(\theta)e_5(ccc-ccs-csc-css-scs-scc+ssc-sss) \\
&\quad + E_3(\theta)e_6(ccc-ccs-csc+css+scs+scc+ssc+sss) \\
&\quad + E_3(\theta)e_7(ccc+ccs+csc-css-scs+scc-ssc+sss) \\
&\quad + e_6\sqrt{\theta_3} \left(\sqrt{\frac{1}{\sqrt{2\pi}}} - 4\theta^{\frac{1}{4}}\sqrt{-\frac{2}{\pi}} \right) e^{-\theta_3}(\cos\theta_3-e_4\sin\theta_3),
\end{aligned}$$

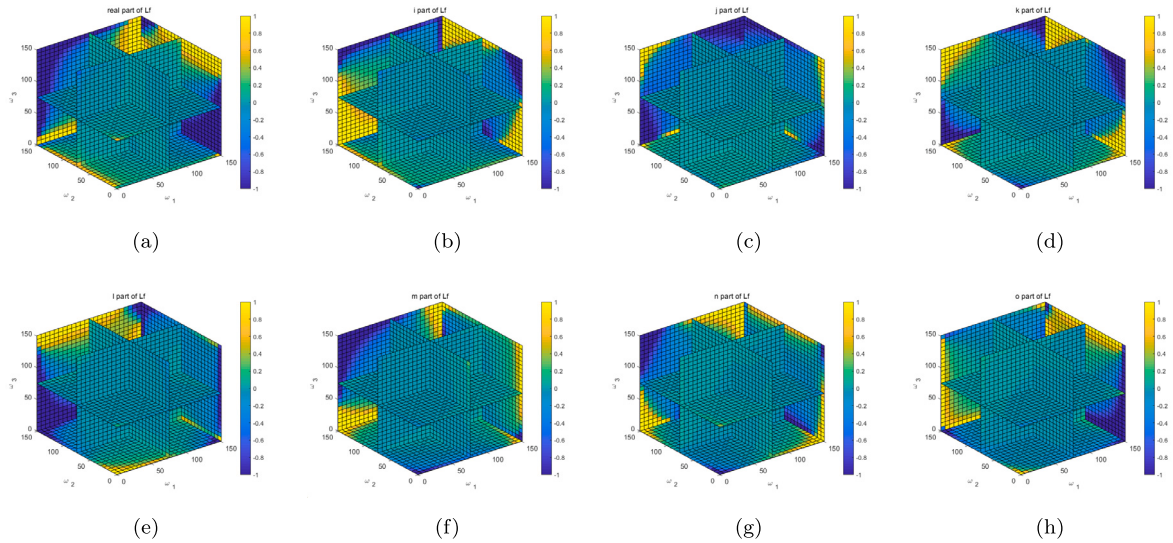


Fig. 9. (a) Real part of $\mathcal{L}_l^o\{f\}(\omega)$; (b) i part of $\mathcal{L}_l^o\{f\}(\omega)$; (c) j part of $\mathcal{L}_l^o\{f\}(\omega)$; (d) k part of $\mathcal{L}_l^o\{f\}(\omega)$; (e) l part of $\mathcal{L}_l^o\{f\}(\omega)$; (f) m part of $\mathcal{L}_l^o\{f\}(\omega)$; (g) n part of $\mathcal{L}_l^o\{f\}(\omega)$; (h) o part of $\mathcal{L}_l^o\{f\}(\omega)$.

$$\begin{aligned}
 I_7 &= \int_{\mathbb{R}^3} \text{Ke} \left[-\left(\frac{a}{2} x_1^2 + \frac{b}{2} x_2^2 + \frac{c}{2} x_3^2 \right) \right] (1 + x_1 x_2 x_3) e_7 dx \\
 &= \frac{1}{\sqrt{e_1 + 1} \sqrt{-e_2 + 1} \sqrt{e_4 + 1}} e^{\left[\frac{\omega_1^2}{2} \left(-\frac{1}{e_1 + 1} \right) \right]} e^{\left[\frac{\omega_2^2}{2} \left(-\frac{1}{-e_2 + 1} \right) \right]} e^{\left[\frac{\omega_3^2}{2} \left(-\frac{1}{e_4 + 1} \right) \right]} \cdot e_7 \\
 &= \frac{e_7}{\sqrt{e_1 + 1} \sqrt{-e_2 + 1} \sqrt{e_4 + 1}} e^{\theta_1 - \theta_2 - \theta_3} e^{e_1 \theta_1} e^{-e_2 \theta_2} e^{e_4 \theta_3} \\
 &= \sqrt{\frac{1}{(-1 - e_1 - e_2 + e_3 - e_4 + e_5 + e_6 + e_7)}} (-1 - e_1 - e_2 + e_3 - e_4 + e_5 + e_6 + e_7) \\
 &\quad e^{\theta_1 - \theta_2 - \theta_3} (\cos \theta_1 + e_1 \sin \theta_1) (\cos \theta_2 - e_2 \sin \theta_2) (\cos \theta_3 + e_4 \sin \theta_3) \\
 &= E_4(\theta) (-ccc - ccs + csc + css - scs - sec + ssc - sss) \\
 &\quad + E_4(\theta) e_1 (-ccc + ccs - csc - css - scs + sec + ssc - sss) \\
 &\quad + E_4(\theta) e_2 (-ccc + ccs - csc + css - scs - sec - ssc + sss) \\
 &\quad + E_4(\theta) e_3 (ccc + ccs - csc + css + scs - sec + ssc - sss) \\
 &\quad + E_4(\theta) e_4 (-ccc + ccs + csc - css + scs - sec - ssc - sss) \\
 &\quad + E_4(\theta) e_5 (ccc + ccs + csc - css - scs - sec + ssc + sss) \\
 &\quad + E_4(\theta) e_6 (ccc + ccs + csc + css - scs + sec - ssc - sss) \\
 &\quad + E_4(\theta) e_7 (ccc - ccs - csc - css - scs + sec - ssc - sss),
 \end{aligned}$$

where

$$\begin{aligned}
 E_0(\theta) &= \sqrt{\frac{1}{(1 - e_1 + e_2 - e_3 - e_4 + e_5 - e_6 + e_7)}} e^{-\theta_1 + \theta_2 + \theta_3} \\
 E_1(\theta) &= \sqrt{\frac{1}{(1 + e_1 - e_2 + e_3 + e_4 + e_5 - e_6 + e_7)}} e^{-\theta_1 + \theta_2 + \theta_3} \\
 E_2(\theta) &= \sqrt{\frac{1}{(-1 + e_1 + e_2 + e_3 - e_4 + e_5 - e_6 - e_7)}} e^{-\theta_1 + \theta_2 + \theta_3} \\
 E_3(\theta) &= \sqrt{\frac{1}{(1 - e_1 + e_2 + e_3 + e_4 + e_5 + e_6 + e_7)}} e^{-\theta_1 + \theta_2 + \theta_3} \\
 E_4(\theta) &= \sqrt{\frac{1}{(-1 - e_1 - e_2 + e_3 - e_4 + e_5 + e_6 + e_7)}} e^{-\theta_1 + \theta_2 + \theta_3}.
 \end{aligned}$$

Remark 5. In the above computation process, we define $s_{(\theta_1)} y_{(\theta_i)} z_{(\theta_j)} = \sin \theta_1 y_{(\theta_i)} z_{(\theta_j)}$, $c_{(\theta_1)} y_{(\theta_i)} z_{(\theta_j)} = \cos \theta_1 y_{(\theta_i)} z_{(\theta_j)}$, and $i = 2, 3$. $y_{(\theta_i)} z_{(\theta_j)} \in \{c, s\}$. The c, s of the above formula denotes the parity function, such as $scs = \sin \theta_1 \cos \theta_2 \sin \theta_3$, $csc = \cos \theta_1 \sin \theta_2 \cos \theta_3$.

7. Discussion

In this section, we discuss and analyze the image characteristics of octonion functions, and examine the corresponding spectral characteristics of the left-sided octonion linear canonical transform (LOCLCT) and left-sided octonion Fourier transform (LOCFT).

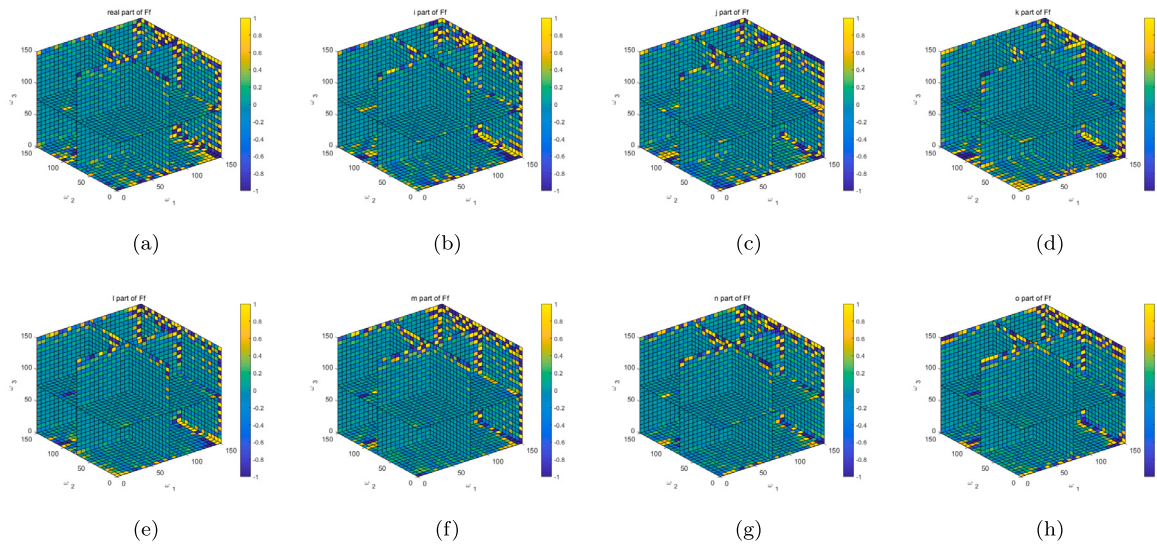


Fig. 10. (a) Real part of $F_l^o\{f\}(\omega)$; (b) i part of $F_l^o\{f\}(\omega)$; (c) j part of $F_l^o\{f\}(\omega)$; (d) k part of $F_l^o\{f\}(\omega)$; (e) l part of $F_l^o\{f\}(\omega)$; (f) m part of $F_l^o\{f\}(\omega)$; (g) n part of $F_l^o\{f\}(\omega)$; (h) o part of $F_l^o\{f\}(\omega)$.

First, from Figs. 5 and 8, it appears that the image in Example 1 looks more regular in space compared to the image in Example 2, which may appear more irregular and less stable.

Second, the spectrum in Example 1, as shown in Fig. 6, is more concentrated, whereas the spectrum in Example 2, as shown in Fig. 9, is more scattered and irregular. The energy distribution in Example 1 is concentrated at certain specific frequencies, whereas the energy distribution in Example 2 is more even or random.

Finally, from the preliminary observations in Figs. 6, 7, 9 and 10, it indicates that the spectrum of LOCFT is symmetric, whereas the spectrum of LOCLCT lacks strict symmetry because of the varying parameters. LOCFT offers a fixed frequency resolution, whereas LOCLCT allows for adjustable frequency resolution by modifying the parameters, enabling a more flexible distribution of the spectrum. This demonstrates that LOCLCT possesses greater flexibility.

The result of the above discussion is not universal. Since the image features of octonion-valued functions are difficult to visualize directly, in our subsequent research, we will further reveal their characteristics and related application scenarios through dimensionality reduction, slicing, parallel coordinate graphs, and other methods.

Next, we present potential applications of LOCLCT. Because the octonion provides more parameters, LOCLCT can be applied to image processing, particularly in the design of color image embedding watermarking algorithms. The octonion transform of the color image allows the embedded information energy to be propagated to all channels simultaneously, rather than just one channel [38]. Other potential applications can be found in image encryption, filter design, pattern recognition, and phase recovery, as mentioned in [39–42].

8. Conclusion

In the present work, the definition of the LOCLCT is presented, and the differential properties of the LOCLCT are studied in detail. A new LOCLCT convolution operator is proposed, and a corresponding convolution theorem is derived. Using these properties and convolution operators, we investigate three-dimensional LTI systems in the LOCLCT domain. Simulations and examples illustrate that the LOCLCT can capture the frequency components and improve flexibility. Additionally, we analyze the spectrum of LOCLCT and discuss its potential applications. Future research will focus on signal reconstruction, image encryption, and other practical applications.

CRediT authorship contribution statement

Nan Jiang: Writing – original draft, Investigation, Conceptualization. **Qiang Feng:** Writing – review & editing, Supervision, Formal analysis. **Xi Yang:** Investigation, Conceptualization. **Jin-Rong He:** Funding acquisition, Formal analysis. **Bing-Zhao Li:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work.

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Data availability

Data will be made available on request.

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